Appendix to **IRS 90940**

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Original

Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol

Appendix A

SFERA Use Cases

This appendix:

- is updated regularly,
- corresponds to the above date of update.

INTERNATIONAL UNION
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Appendix A - SFERA Use Cases

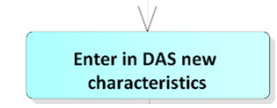
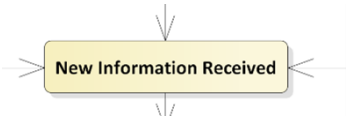
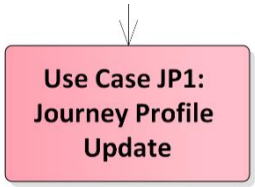
This appendix provides a detailed description of the use cases presented in Chapter 4 of the IRS. The description includes activity diagrams and – where applicable – sequence diagrams that show the flow of information and data. The sequence diagrams contain the names of the SFERA messages used, which are described in detail in Appendix E.

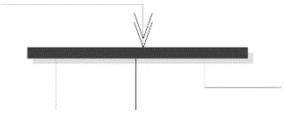
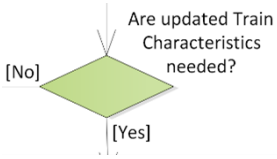
The use cases are classified by the following topics:

1. DX (Data Exchange): use cases related to the connection and disconnection process, and the transition between IM networks;
2. JP (Journey Profile): use cases related to the transmission of the elements necessary for a DAS to generate advice in a C-DAS-O configuration (JP and associated elements TC, SP);
3. TC (Train Characteristics): use cases when a driver declares an error or a change in the Train Characteristics used at a given moment;
4. PW (traction PoWer): specific use cases related to the management of the power (positive or negative) used by trains on electrified routes, and the traction power infrastructure (substations, distribution grid);
5. SR (Status Report): use cases related to the exchange of information initiated by the DAS, in majority for feedback purposes to the TMS;
6. DM (Driving Mode): use cases related to the management of the driving mode of different DAS devices;
7. CDC (C-DAS-C): use cases related to the transmission of the elements necessary for a DAS to present advice in a C-DAS-C configuration.
8. AF (Additional Functions): use cases related to additional functions.

The last section presents an example of how the sequence diagrams can be extended to the IM-RU setup.

The activity diagrams have been described using UML. The following table is the legend for these diagrams:

Element	Meaning
	Human or automatic action
	Automatic action
	Reference to another use case

Element	Meaning
	Parallel actions
	Point of entry
	Point of exit
	Conditional paths

1 DX: Data Exchange Messages

1.1 DX1: Connection Handshake

This use case presents how a SFERA connection is established between a receiver and a provider on a functional level.

Before any functional level exchanges can take place, the DAS-OB shall connect to the DAS-TS on a technical level. This is described in Appendix B (SFERA Communication Protocol).

Once this level has been established, the DAS-OB will send a handshake request, followed by an acknowledgement message by the ground. Those messages will include functional parameters to be used in the subsequent use cases (e.g. timeouts and available functions on both sides, as well as information on related trains - the ‘own train’, downline and upline trains). Notable parameters concern the interaction modes. The DAS-OB will send the possible triplets of operation:

- Driving Mode (Inactive, Read-only, Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4)
- Architecture (DAS-O or DAS-C)
- Connectivity (Connected or Standalone)

The IM will respond, accepting or rejecting the handshake. If the handshake is accepted, the accepted pair (Architecture, Connectivity) to be used is sent to the DAS-OB.

A handshake request can be rejected for several reasons (e.g. if the on-board and trackside ATO or SFERA versions are incompatible, if the trackside ATO or DAS is not in charge for the particular journey, or if the requested architecture, connectivity and/or driving modes are not supported).

A transition between DAS-C and DAS-O, or from standalone to connected would need a new handshake.

Once this exchange has been performed successfully:

- in DAS-C operation, the DAS-OB can request C-DAS-C advice for this train service to be able to start generating advice.

- in DAS-O operation, the DAS-OB can request the JP for this train service to be able to start generating advice.

This use case is described in the activity diagram shown in Figure 1 and in the sequence diagram shown in Figure 2. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

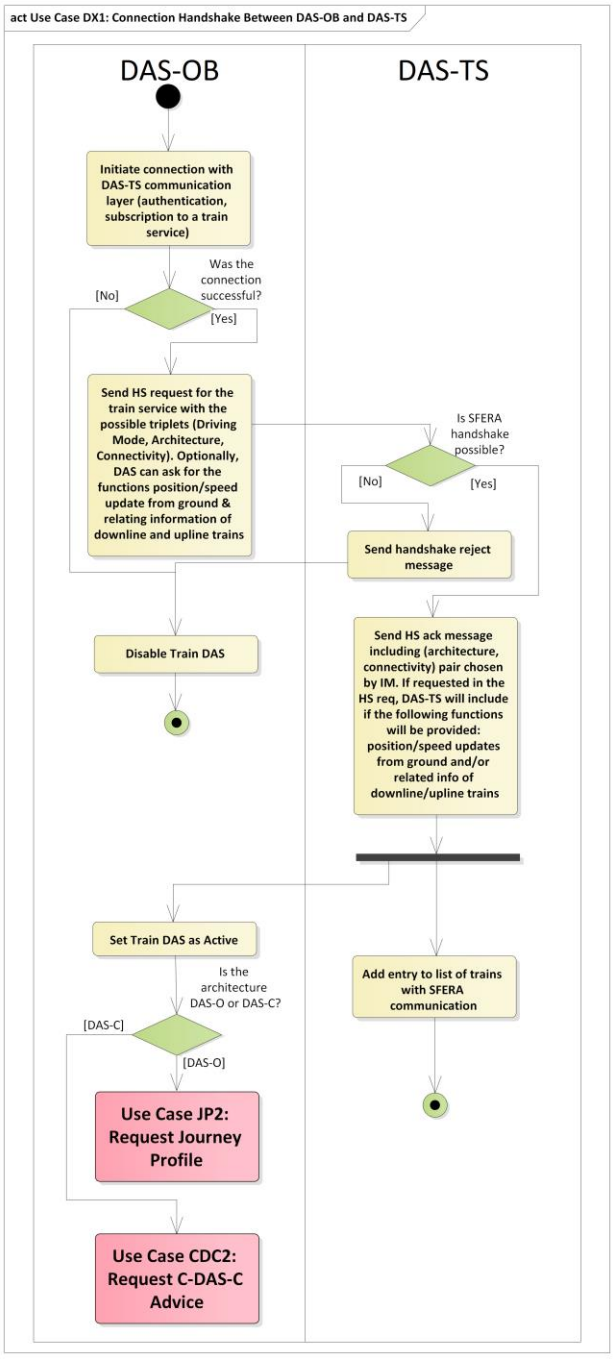


Figure 1: Activity Diagram – DX1

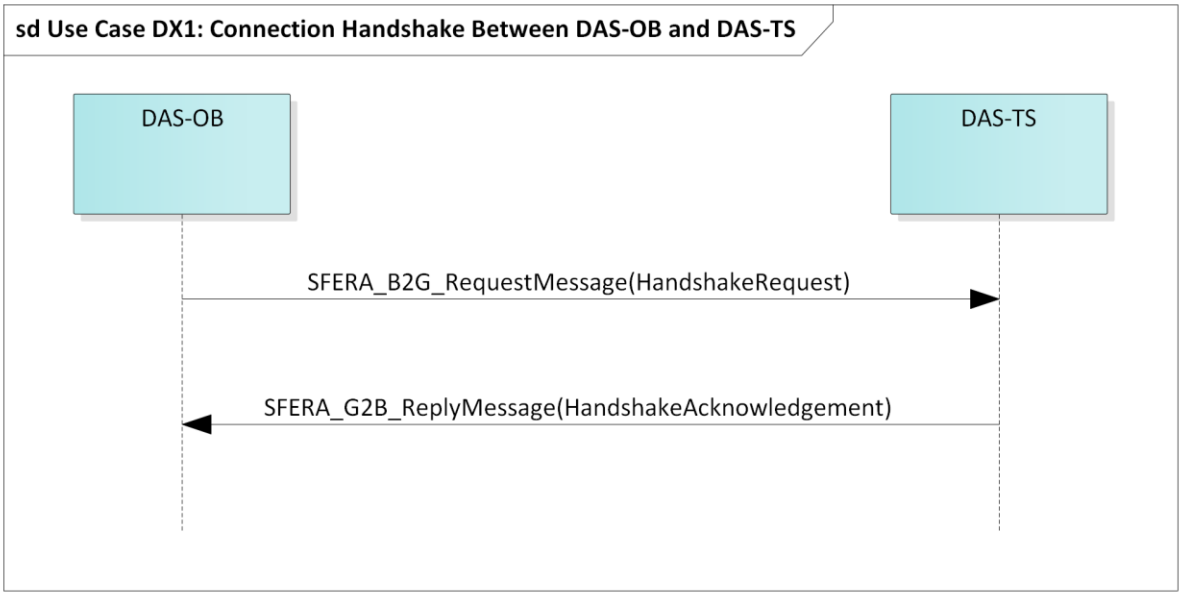


Figure 2: Sequence Diagram – DX1 (if handshake is possible)

1.2 DX2: Session Termination from DAS-OB

In SFERA, it is considered preferable to use a functional session termination before the DAS-OB closes the communication layer with the DAS-TS. This means that the ground servers will be able to differentiate between a DAS-OB having lost its connection or ending operation in a controlled manner (e.g. driver handing over responsibility for the train).

This use case is described in the activity diagram shown in Figure 3. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

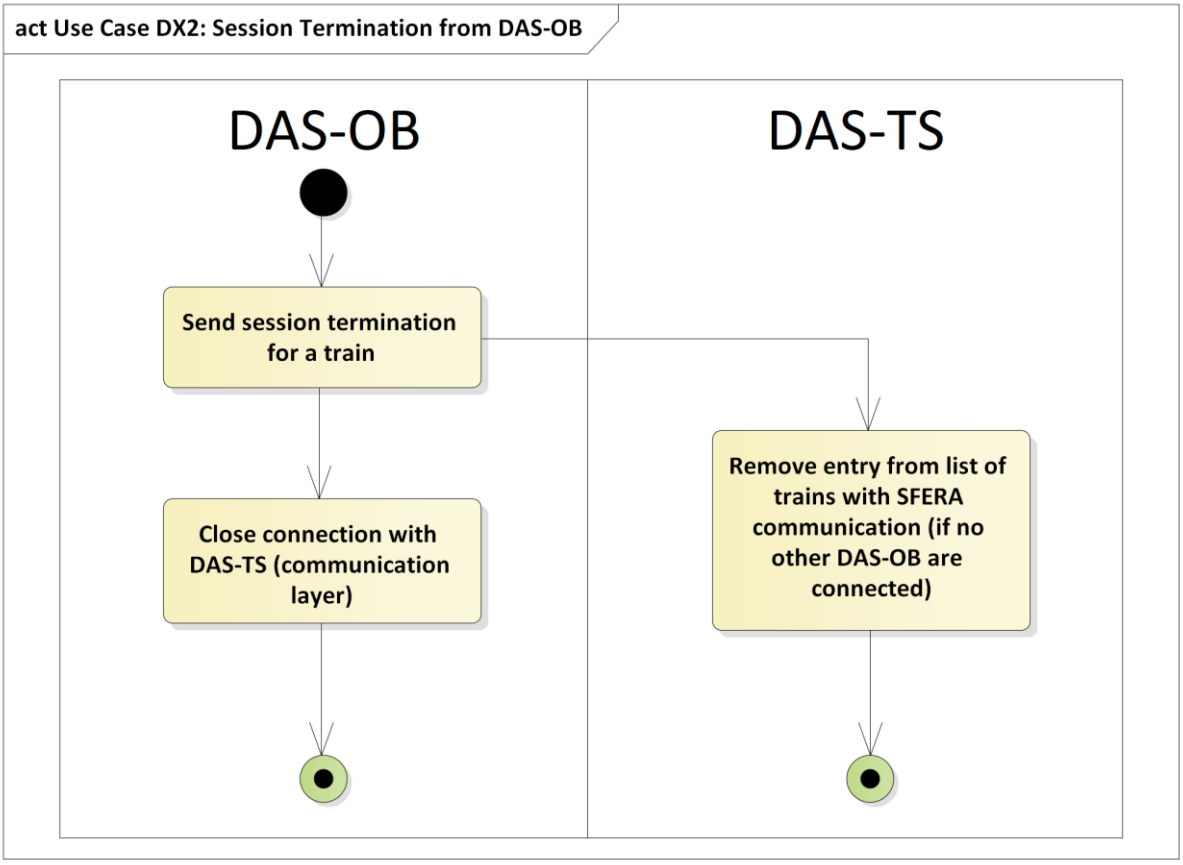


Figure 3: Activity Diagram – DX2

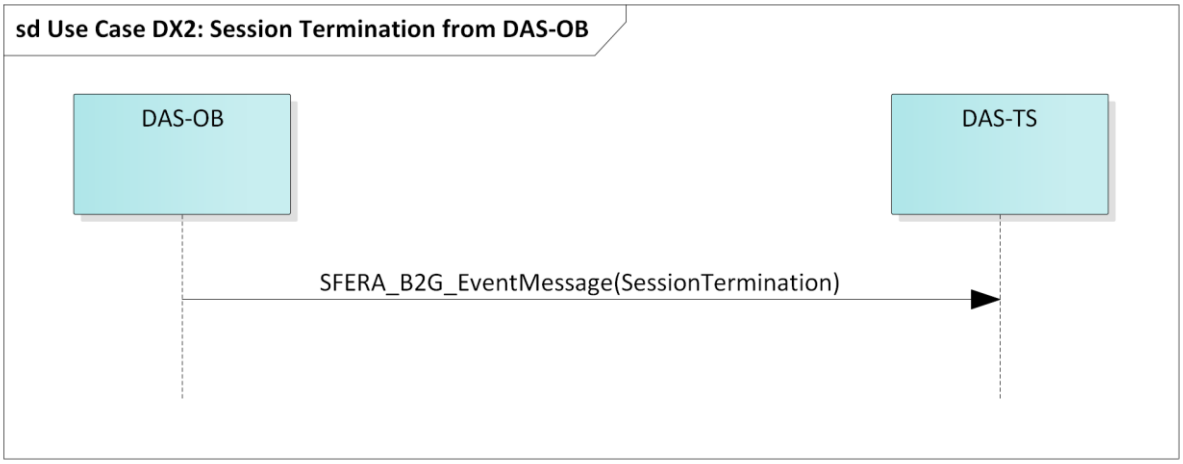


Figure 4: Sequence Diagram – DX2

1.3 DX3: Handover between two IM DAS-TS

This use case describes the situation when a train journey crosses a border between two different DAS-TS zones (from DAS-TS1 to DAS-TS2).

Before the border, at a moment when DAS-TS2 has opened access to the data for the considered train, the receiver can open the connection. The DAS knows which is the next DAS-TS to interrogate, because the last SP referenced by the JP received from DAS-TS1 shall include this information (identifier of the next IM to interrogate). This information can also be included in C-DAS-C advice. The IM policies of DAS-TS2 shall aim to permit the DAS to connect and obtain information before leaving its last stop before the border

(functional errors can arise if a DAS does not have information from its current position to the next point it will stop at). For trains leaving several hours before the border without any intermediate stops, the time of availability of the data shall be taken into account.

When the train has passed the border, it can disconnect from DAS-TS1, and from then on only maintain the connection with DAS-TS2.

This use case is described in the activity diagram shown in Figure 5. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

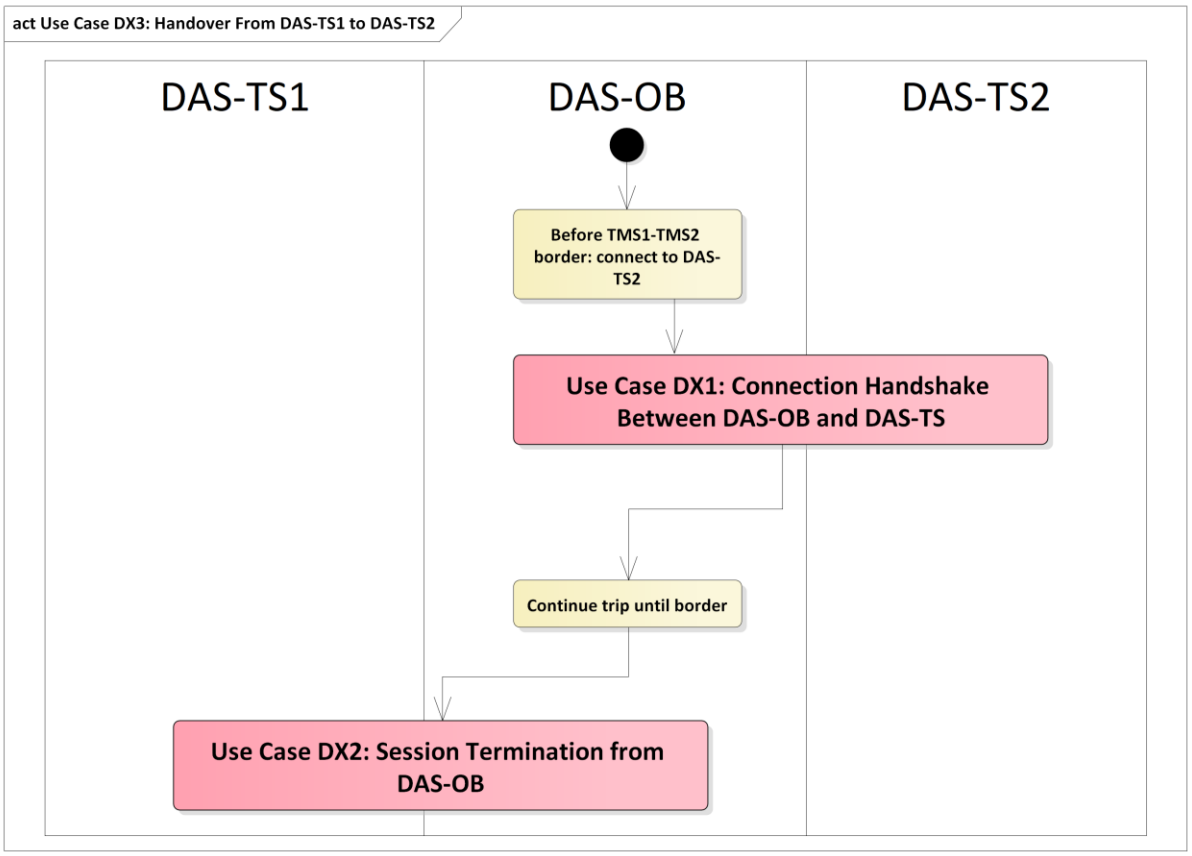


Figure 5: Activity Diagram – DX3

1.4 DX4: Session Termination Request from DAS-TS

In some cases, the DAS-TS will terminate the connection with the DAS-OB. The DAS-TS will send a session termination request to the board. This will then lead to DX2, which is supposed to introduce a session termination message from the DAS-OB to the DAS-TS. If this is not the case, the DAS-TS will monitor a timeout delay for receiving this session termination and the SFERA communication will be considered terminated.

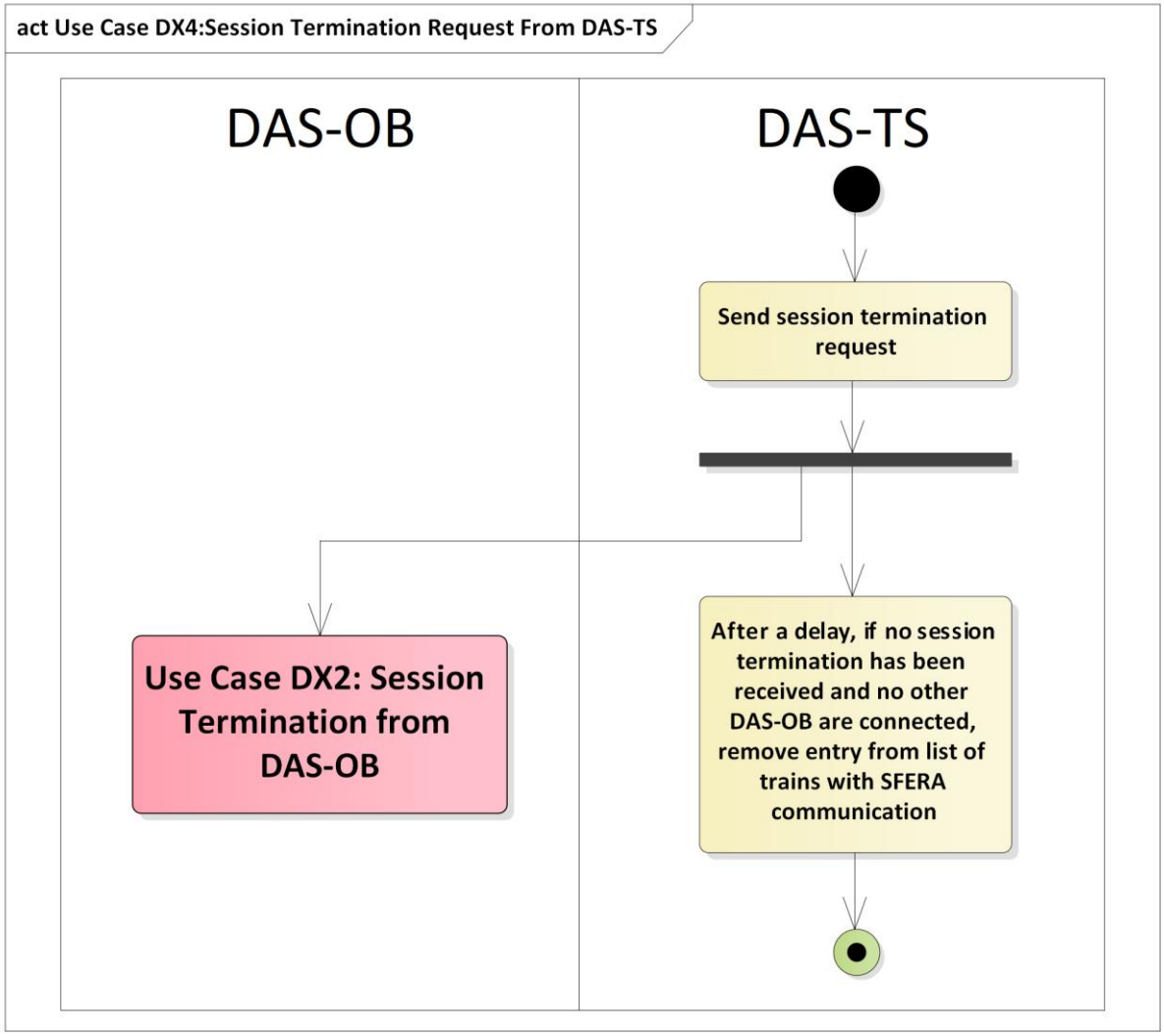


Figure 6: Activity Diagram – DX4

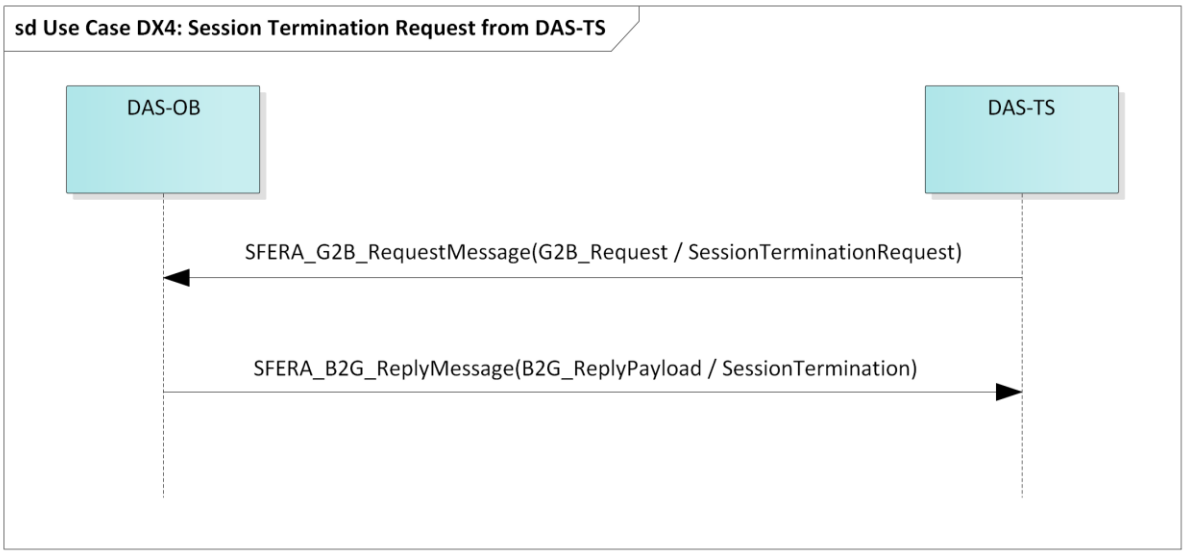


Figure 7: Sequence Diagram – DX4

This use case is described in the activity diagram shown in Figure 6. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

2 JP: Journey Profile Messages (and related)

2.1 JP1: Journey Profile Update

This is the main use case in C-DAS mode, describing how updates of the real time traffic plan are sent to the DAS while the DAS is active, with an active connection to the DAS-TS.

Information generated internally (e.g. by dispatcher decisions), or externally from the trains or other sources, will be received by the TMS. The TMS will use this information to trigger updates of the real-time traffic plan. If the update impacts the Journey Profile of one or more trains, the JP3 Use Case is triggered for all these trains.

This use case is described in the activity diagram shown in Figure 8. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation (DAS-O)

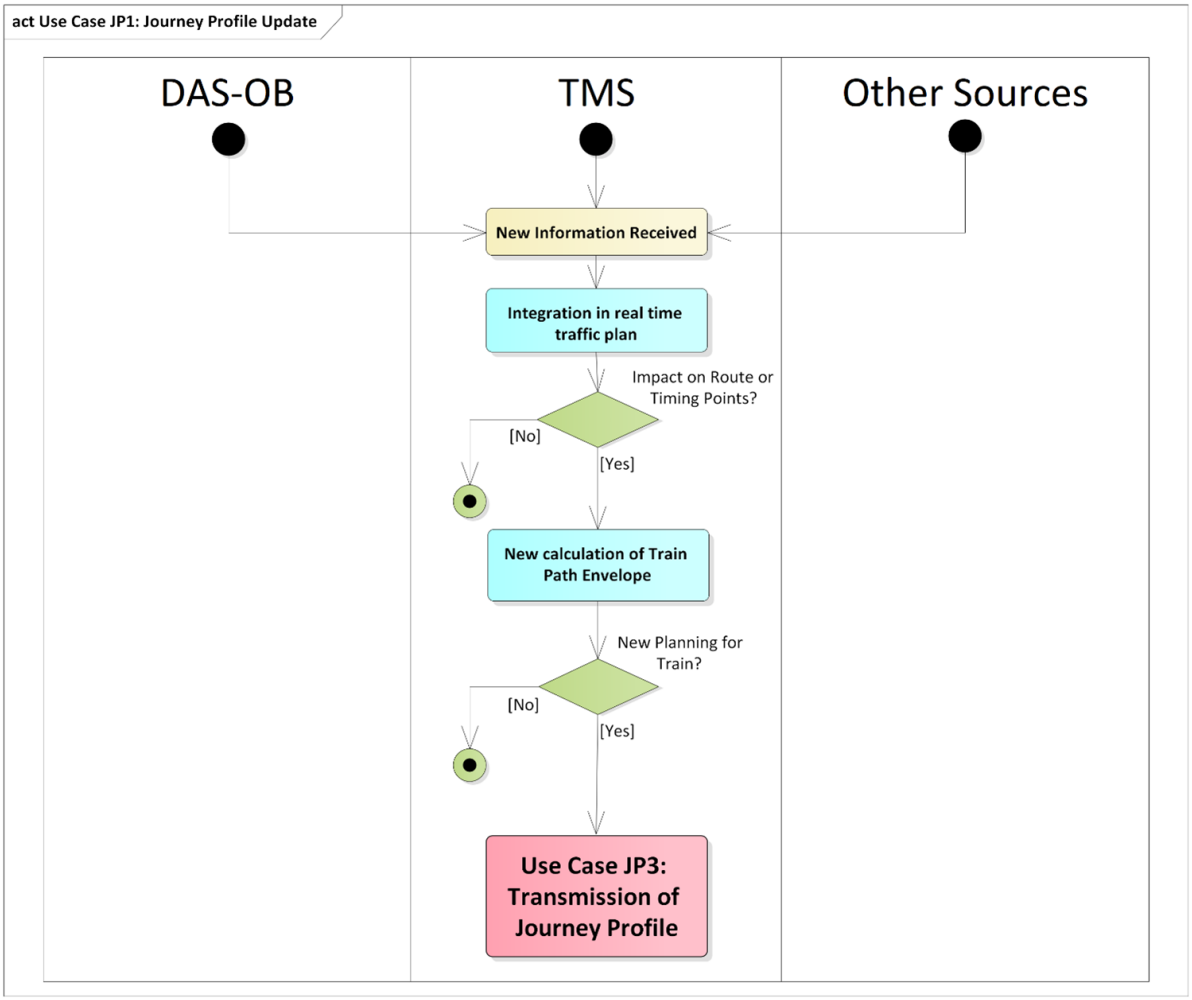


Figure 8: Activity Diagram – JP1

2.2 JP2: Request for Journey Profile

This is the core use case for an S-DAS, triggered once the SFERA connection has been established. It will also be used by a C-DAS upon initialisation and in cases when the DAS needs to check for updates; for example, when reconnecting after a journey portion where the on-board device has lost connection with the ground.

If the receiver requests Journey Profile from the provider without specifying the version of the JP that is in use (e.g. upon first connection), one of three outcomes can be expected:

- If the data is available for the train, JP3 is triggered.
- If the data is not available at this time, but will be available later, an error message is sent which invites the receiver to try again later.
- If the provider does not provide information for this train (now or later), an error message is sent, which terminates the connection.

If the receiver has provided the version of the JP in use, and the provider does not have a more recent version of the JP available, an additional outcome is possible: a simple acknowledgement.

This use case is described in the activity diagram shown in Figure 9 and in the sequence diagram shown in Figure 10. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	BoardAdviceCalculation (DAS-O)

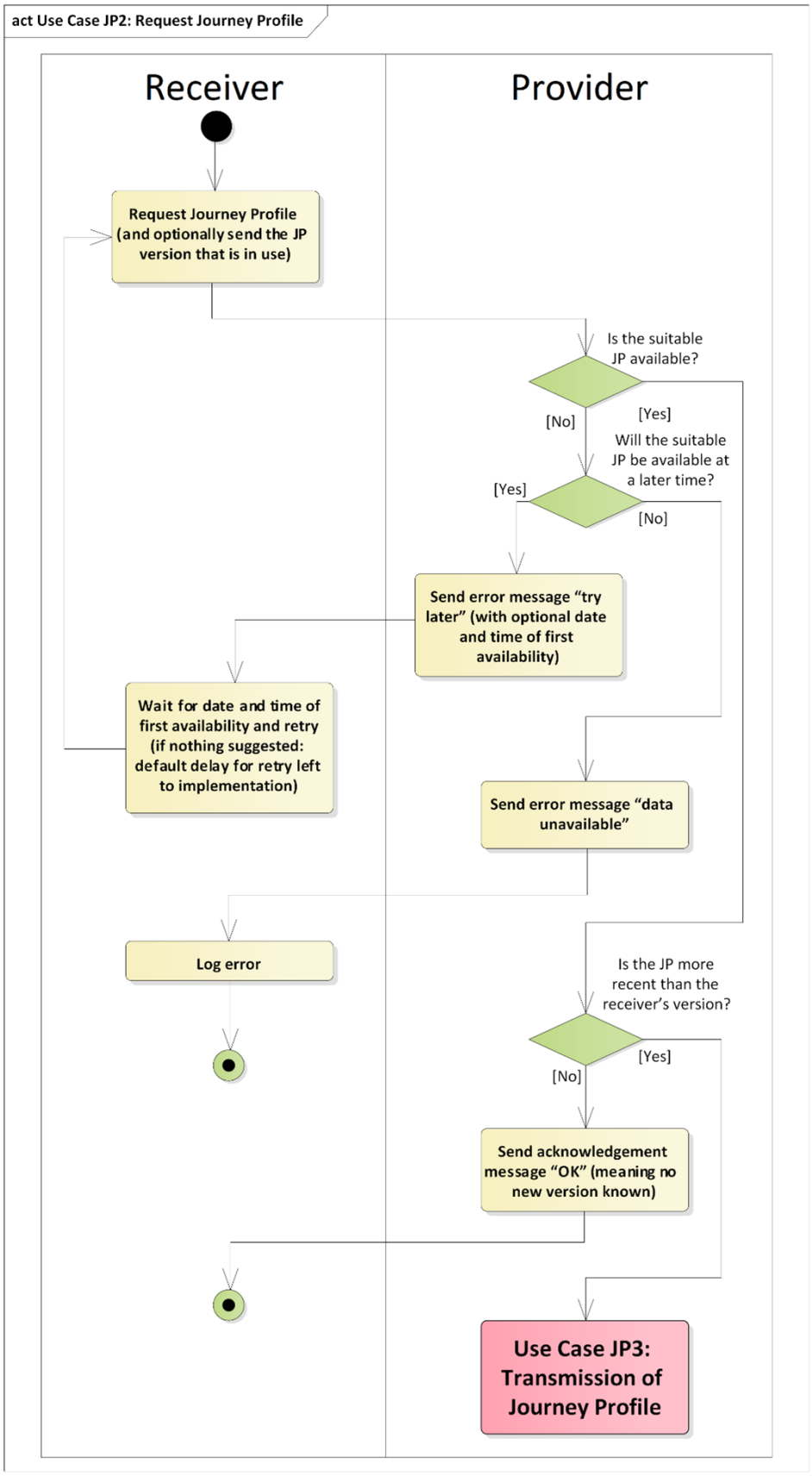


Figure 9: Activity Diagram – JP2

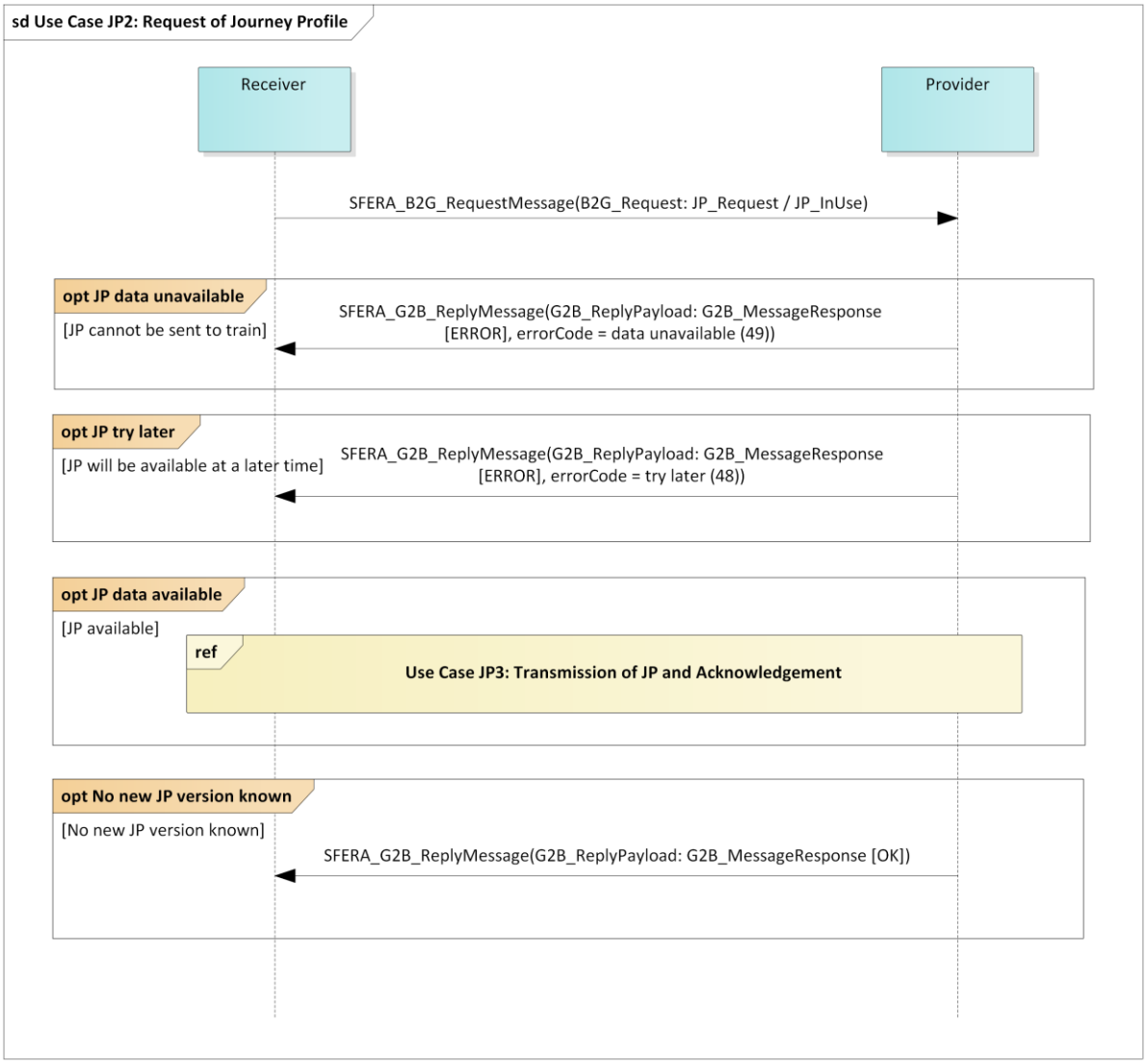


Figure 10: Sequence Diagram – JP2

2.3 JP3: Transmission of Journey Profile and Acknowledgement

JP3 presents the sequence of actions used to ensure that the receiver gathers a complete set of information upon receiving a new Journey Profile. The JP can reference SPs and/or TCs not known by the receiver (either new ones or those with updated version numbers); the receiver checks its cache (if any) if it already has this version of the SP or TC. If this is the case, Use Cases JP4 and/or JP5 will be used. These use cases can be applied in parallel.

If the JP received could not be processed correctly by the receiver, an error message is sent to the provider.

The receiver confirms that the JP has been received in full (but not necessarily processed and actively used by the DAS-OB) by sending an acknowledgement message to the provider.

After receiving the updated JP, the on-board DAS verifies whether it can respect the new Timing Points specified by the JP. If not, it shall communicate to the TMS that the time window cannot be respected via Use Case SR2. If the DAS-OB has been able to process the JP and define a driving profile, a Status Report will be sent to the TMS, specifying the new version of the JP being used (JP_InUse attribute). Only after receiving this Status Report can the TMS consider that the train will take into account the new objectives.

This strategy is different from the one defined for ERTMS/ATO:

- In ERTMS/ATO (using SUBSET-126 and the associated communication protocol), the sender is responsible at a functional level for resending a message if no acknowledgement has been received from the DAS-OB.
- In the SFERA communication layer (for DAS-OB/DAS-TS links), a guaranteed delivery mechanism is implemented, meaning that once the message has been sent, it is supposed to arrive at its destination, even if the DAS-OB is in a “GSM white zone” at the moment it is sent (the message will be kept in the communication channel until the DAS is back online). If the provider in SFERA were to adopt the same strategy as ERTMS/ATO in the event of a loss of connection, the communication channel would accumulate a backlog of identical messages, which would all be sent together upon reconnection.

This use case is described in the activity diagram shown in Figure 11 and in the sequence diagrams shown in Figure 12 and Figure 13. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

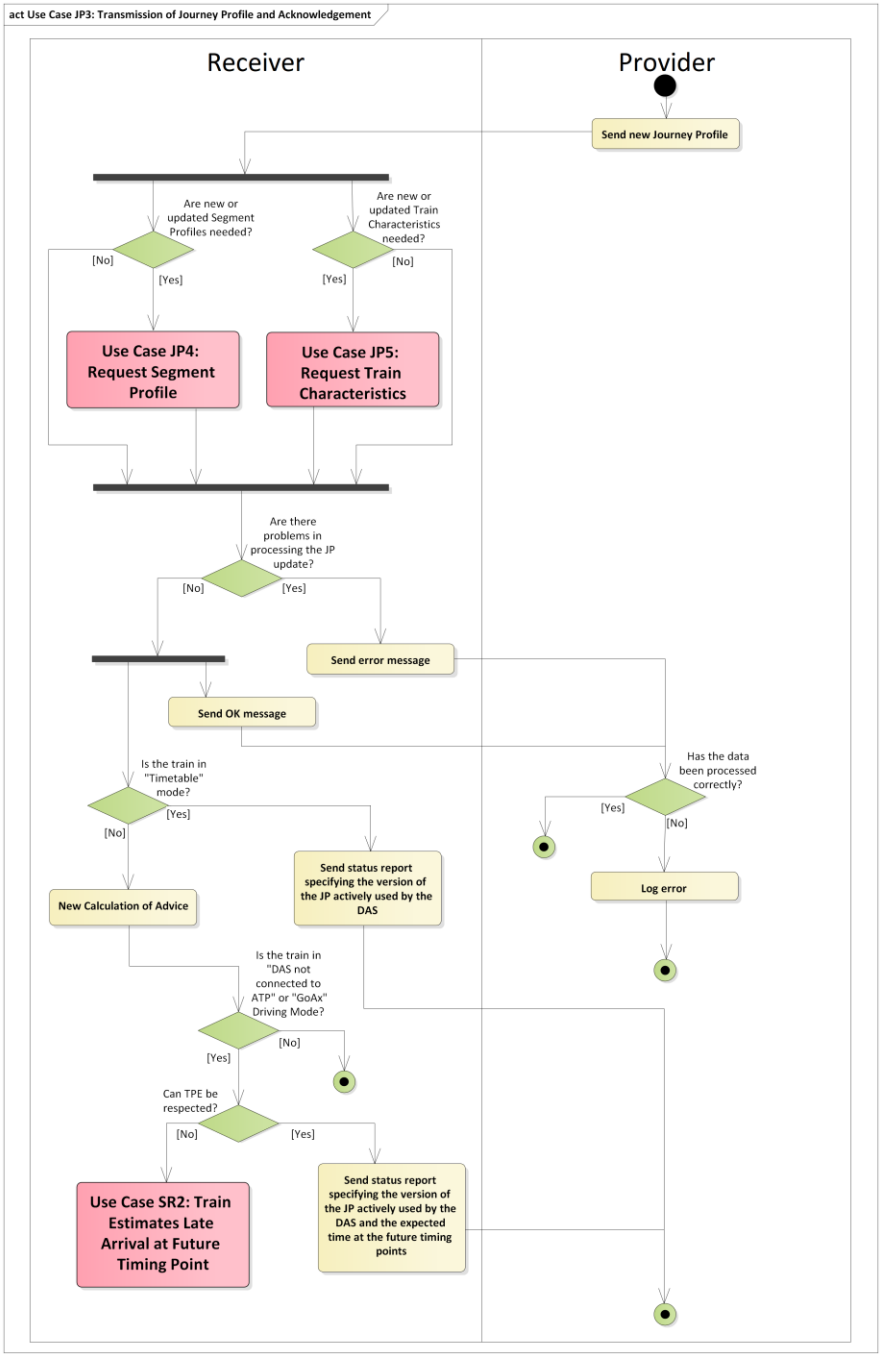


Figure 11: Activity Diagram – JP3

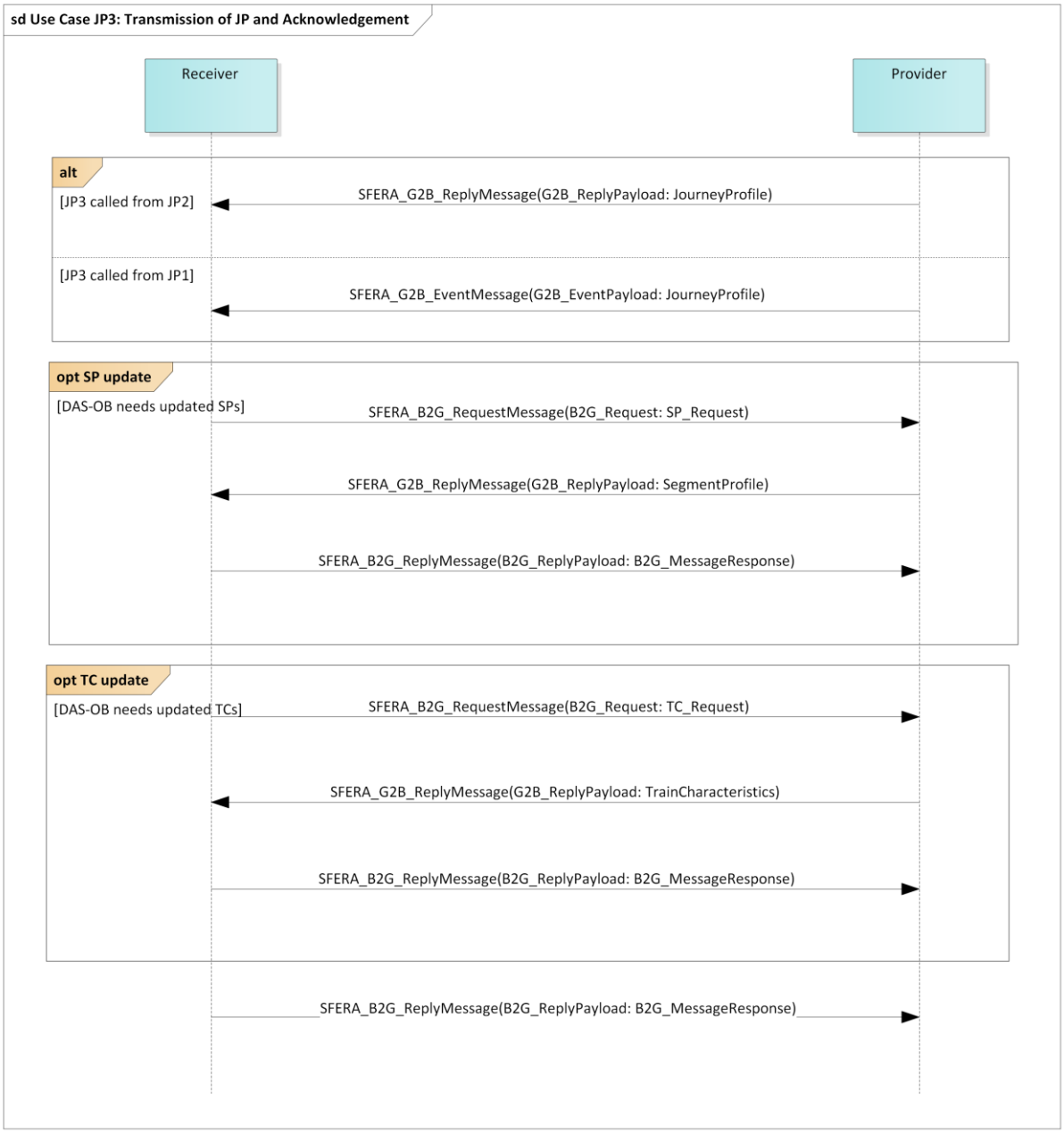


Figure 12: Sequence Diagram – JP3 (JP, TC and SP transmission)



Figure 13: Sequence Diagram – JP3 (Status Report)

2.4 JP4: Transmission of Segment Profile and Acknowledgement

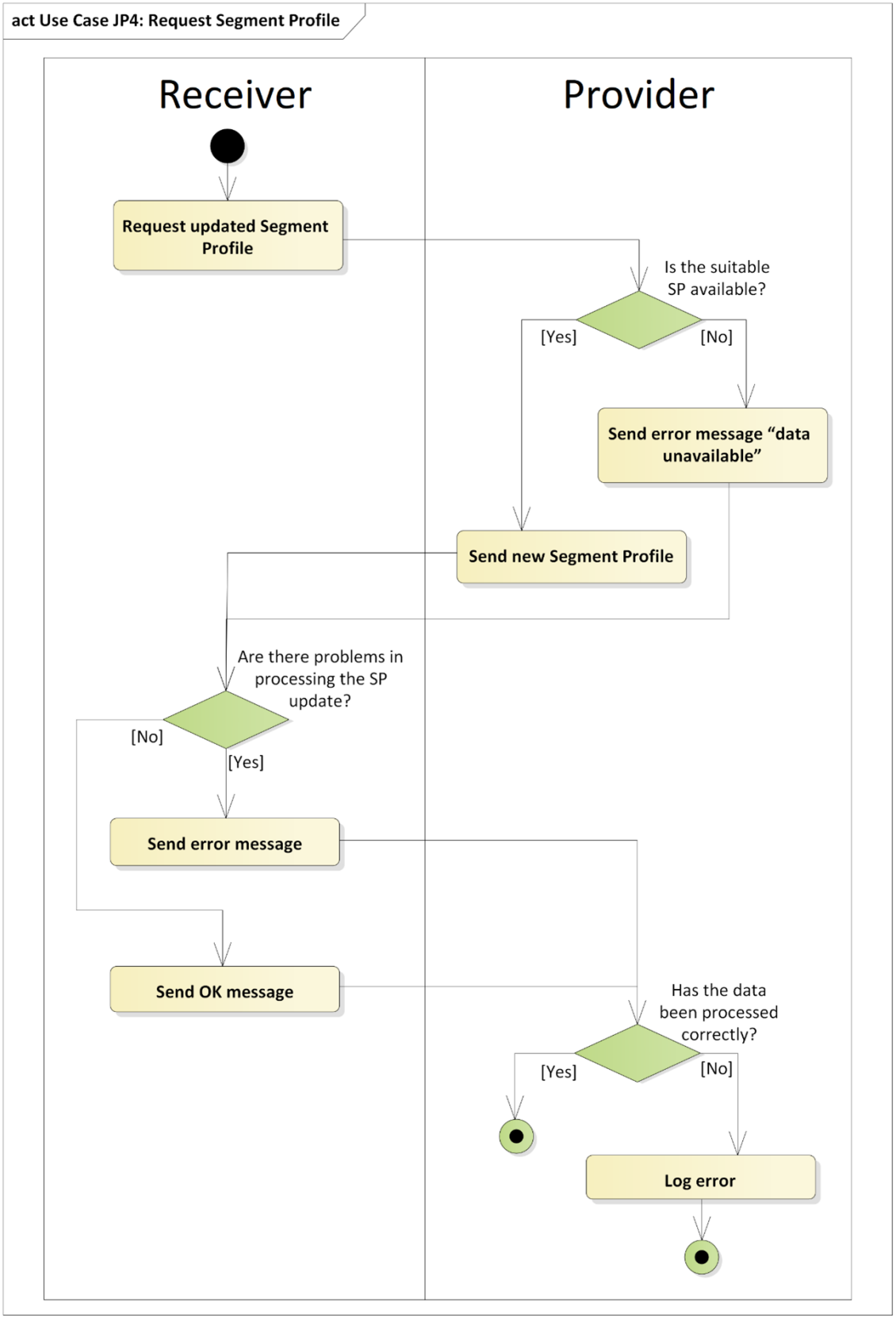


Figure 14: Activity Diagram – JP4

This use case describes the error handling to be implemented when an SP is sent from the provider to the receiver. The error handling is similar to the error handling in transmitting Journey Profiles (JP3).

This use case is described in the activity diagram shown in Figure 14. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	All

2.5 JP5: Transmission of Train Characteristics and Acknowledgement

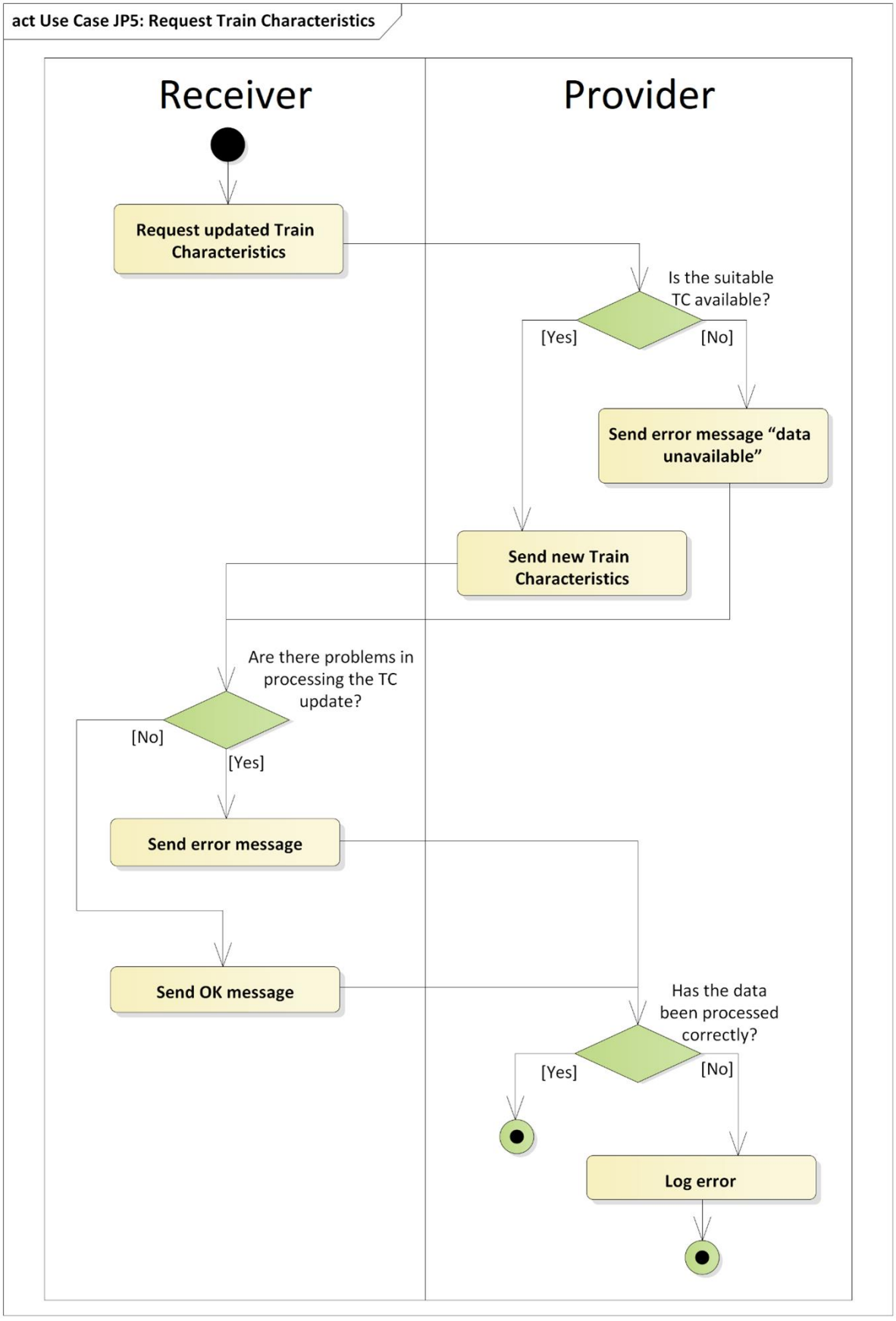


Figure 15: Activity Diagram – JP5

This use case describes the error handling to be implemented when a TC is sent from the provider to the receiver. The error handling is similar to the error handling in transmitting Journey Profiles (JP3).

This use case is described in the activity diagram shown in Figure 15. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	All
DAS Architecture	BoardAdviceCalculation(DAS-O)

3 TC: Train Characteristics Change from Board to Ground

SFERA allows for some Train Characteristics (e.g. train length, weight, maximum speed, degraded acceleration or braking, etc.) to be changed by the driver and transmitted to the ground. Such changes will result from two main cases:

- The initial TAF/TAP declaration of the RU is not coherent with the real train used by the driver.
- The right train has been declared, but the train has degraded functions.

TC1 and TC2 describe two ways of handling the correction of the Train Characteristics associated with the JP. This correction can be declared manually by the driver, or automatically if the DAS is connected to the TCMS.

Both use cases integrate common assumptions:

- The DAS immediately takes into account the correction to generate advice.
- The information is sent as soon as possible to the TMS, which can choose to use the information for non-safety-related functions.
- The information is sent to the RU back office (by computer message, or by telephone), so that the RU can send an updated TAF/TAP message to the TMS (reference for the TMS for safety-related functions: train length, etc.).

The possibility of sending the information before the TAF/TAP message is generated has been introduced for the following reasons:

- The process of generating the TAF/TAP message can take some time, so it can be useful to start using the new data in the TMS before receiving the TAF/TAP message.
- SFERA has additional information compared to the TAF/TAP message.

When the driver enters new characteristics, the DAS will identify if it needs to interrogate the ground to get new Train Characteristics. This may not be necessary if, for instance, the DAS has pre-loaded TCs for alternative rolling stock, or if it is an integrated DAS which can get this information from the TCMS. If not, the DAS will apply JP5. By doing this, the train is capable of delivering advice before the TMS has had time to process the TC change and generate a new JP.

While the Train Characteristics data structure (see Appendix D and E for details) allows the transmission of a significant amount of data when performing a Train Characteristics

change, it is always possible to send a message changing only a limited number of features (e.g. just the train length or weight).

3.1 TC1: Train Characteristics Change (Train notifies TMS and RU)

In this use case, the information is sent in parallel to the RU and/or the TMS. The information sent directly to the TMS will use the SFERA protocol. Concerning the information sent to the RU back office, two choices are possible:

- In the “IM-RU” setup, the RU back office intercepts the TC change message to trigger an action from the rolling stock manager to trigger a new TAF/TAP message;
- In the “IM-Train” setup, the rolling stock manager is informed:
    - by a specific message from the DAS to the RU;
    - by a telephone call from the driver.

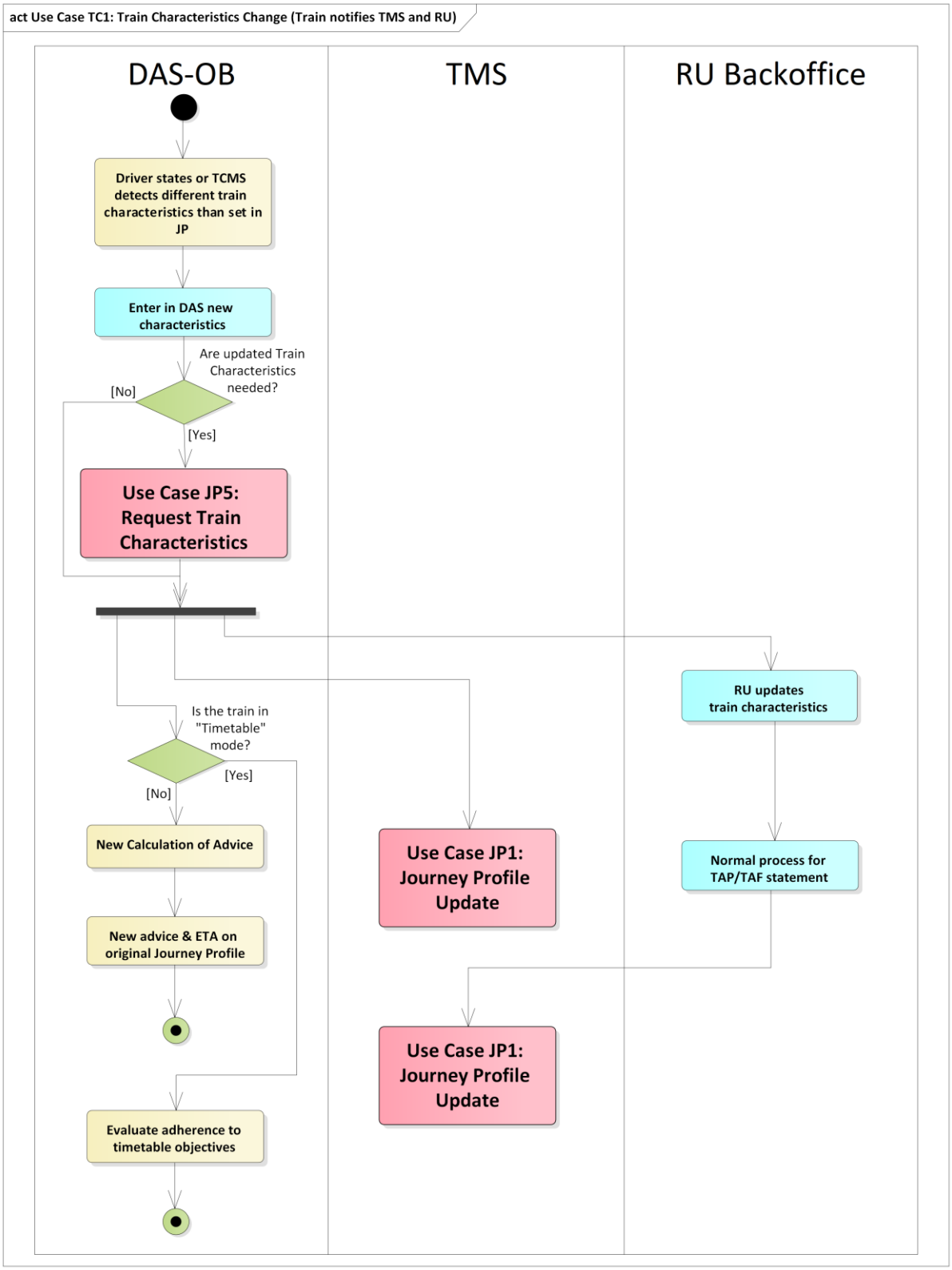


Figure 16: Activity Diagram – TC1

This use case is described in the activity diagram shown in Figure 16 and in the sequence diagram shown in Figure 18. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)

3.2 TC2: Train Characteristics Change (Train notifies RU which notifies TMS)

In this use case, the TC change systematically goes through the RU back office to be verified by the rolling stock manager. Once the manager has verified the information from the driver, the TMS is notified as soon as possible to allow its use for non-safety-related functions and for sending information not covered by the TAF/TAP message. The TAF/TAP message generation procedure is performed in parallel and sent to the IM to formalise the change and notably for safety-related functions.

This use case is described in the activity diagram shown in Figure 17 and in the sequence diagram shown in Figure 18. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)

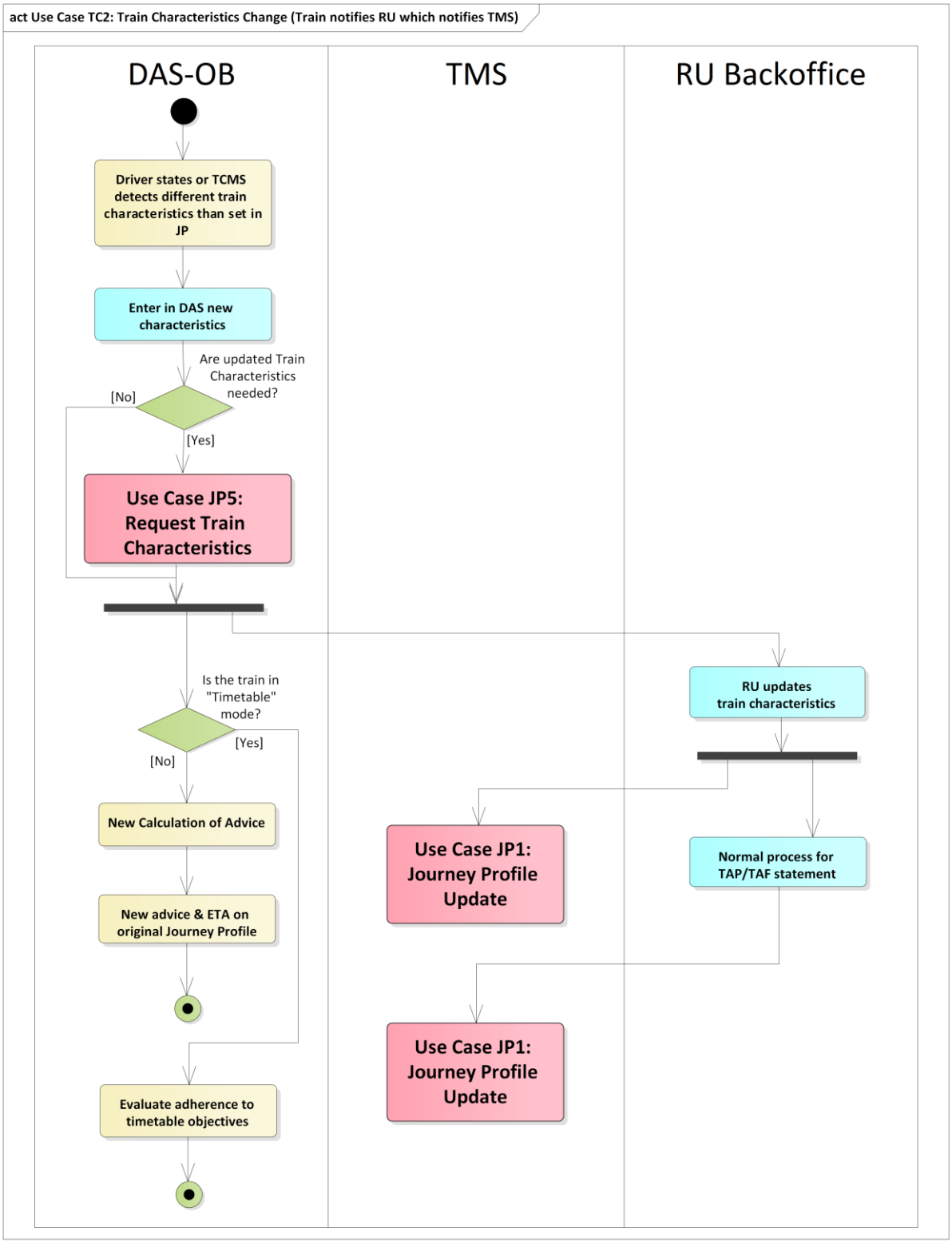


Figure 17: Activity Diagram – TC2

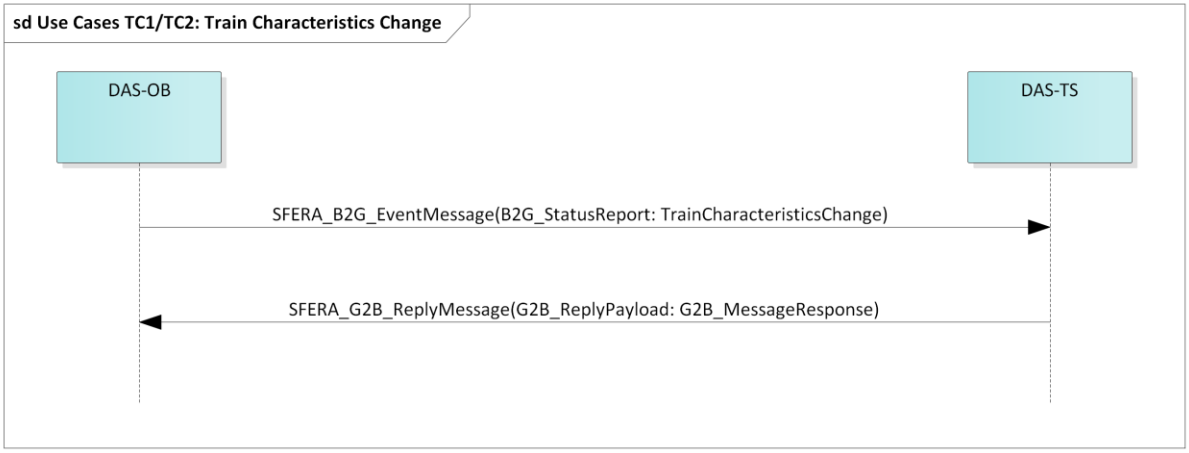


Figure 18: Sequence Diagram – TC1 and TC2

This sequence diagram presents the messages exchanged specifically for this use case. The sequence diagrams for JP5 and JP1 can be found in their respective chapters.

4 PW: Traction Power

4.1 PW1: Manage Traction Power Limitations on Trains

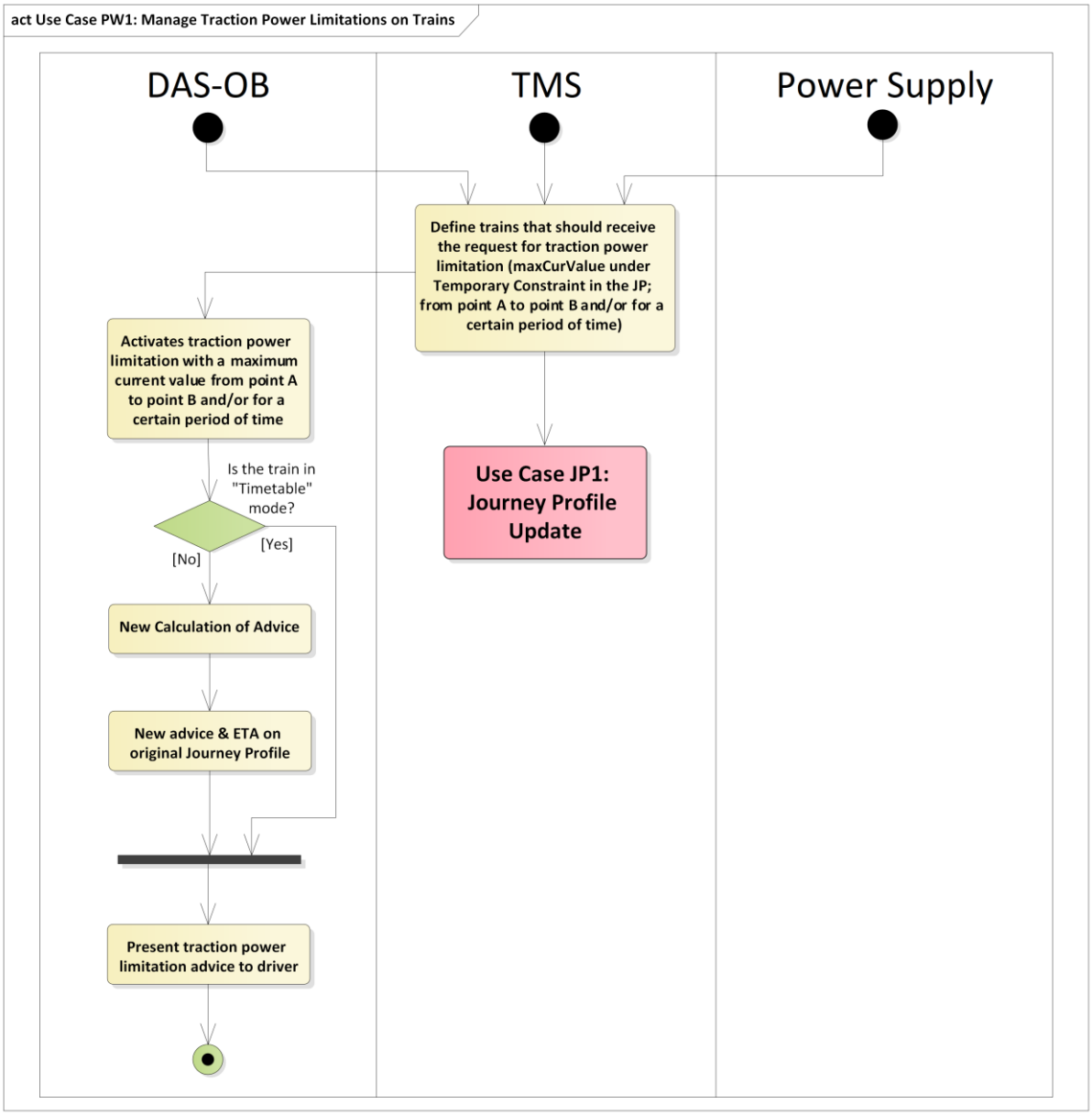


Figure 19: Activity Diagram – PW1

The basic “traction power limitation” use case (named Use Case PW1) involves the collection by the TMS of power information from all trains and the power supply, and the calculation of whether there has to be a traction power limitation request to a train. The JP needs to be updated to take this limitation into account.

The aim is to reduce the current taken from the Overhead Contact Line. Obviously the driver can also choose not only to limit the current used for traction but also to reduce the auxiliary current (e.g. for HVAC).

This use case is described in the activity diagram shown in Figure 19 and in the sequence diagram shown in Figure 20. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All

Operational Mode	Contexts in which this use case is applicable
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)



Figure 20: Sequence Diagram - PW1

4.2 PW2: Manage Traction Power Limitations (due to external causes)

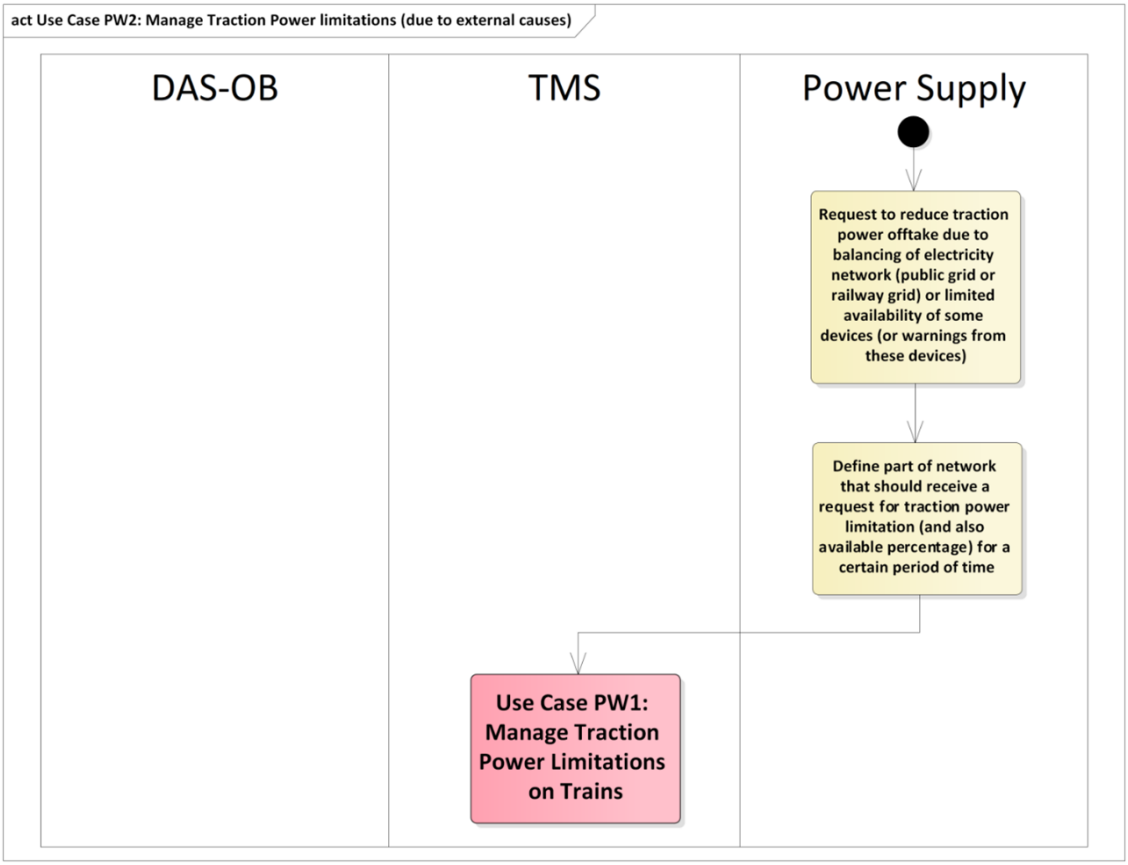


Figure 21: Activity Diagram – PW2

Use case PW2 combines many external causes that can result in a request to reduce the traction power in a certain period of time and for a specific area. These goals can be technical or economical:

- reducing costs for own electricity production;
- reducing costs of the electricity bill by reducing the highest power peaks and therefore the subscribed power of the substation;
- avoiding high investment costs in reserve capacity for the railway electricity grid (with less components in parallel, a component failure might cause current limitations for some trains);
- keeping the traction grid alive in case of a critical component failure or a thermal alarm;
- using thermal and kinetic inertia of railways to support the stability of the European electricity transmission grid through demand-response mechanism.

The last goal can result in an implicit decreasing capacity reserve obligation or in an explicit offering capacity reserve directly to the Transmission System Operators or, more likely, via an aggregator combining the possibilities of different end users. The railway sector can offer a strong reduction of power in a varying period. Other actors (possibly also in the railway system) should be able to scale up production or cut other consumers.

Short response times are needed to reduce high peaks and support the stability of the grid. This might be a specific implementation. Avoiding the switch-off of parts of the electricity grid due to overcharging of lines or components is also possible with longer response times.

This use case is described in the activity diagram shown in Figure 21. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)

4.3 PW3: Manage Traction Power Limitations (due to dense traffic)

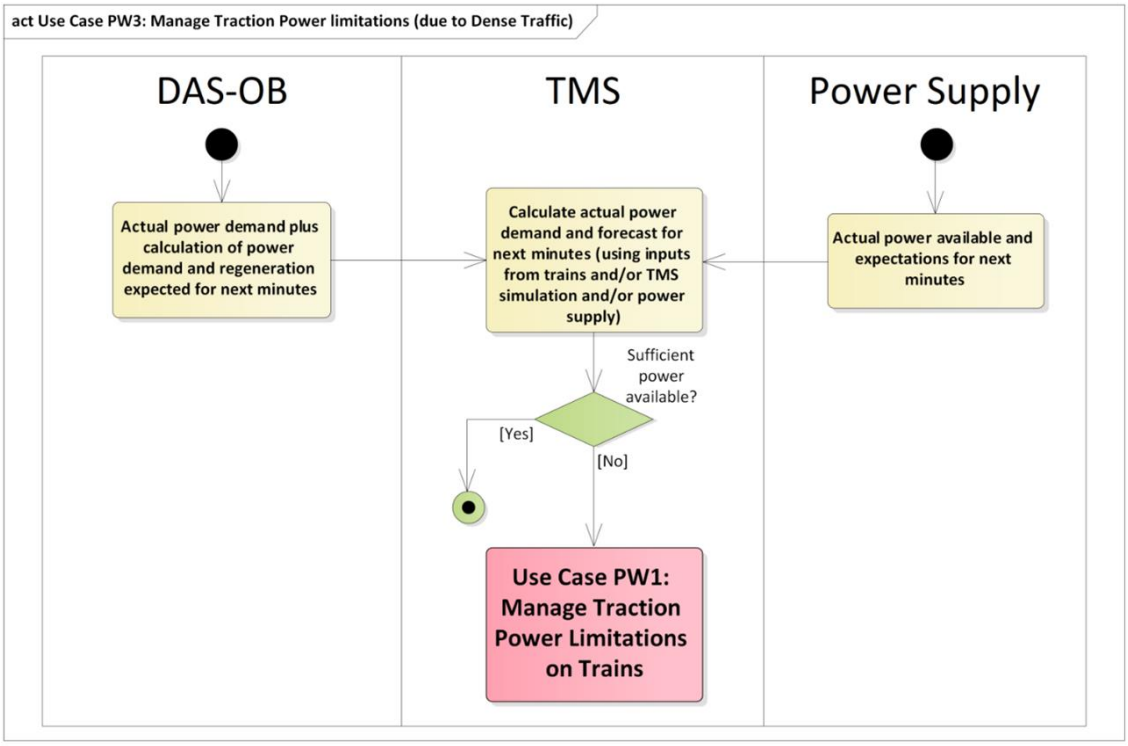


Figure 22: Activity Diagram – PW3

In this use case, the TMS calculates the actual power demand and compares this with the actual available power. This calculation is done for the next minutes. It might also use simulations made by the TMS.

This use case also permits an increase in the capacity of existing tracks without investing in extra available power. The static current limitations applicable to all trains on a part of the traction grid are replaced by dynamic current limitations only applicable when needed.

This use case is described in the activity diagram shown in Figure 22 and in the sequence diagram shown in Figure 24. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)

4.4 PW4: Optimise Regenerative Braking

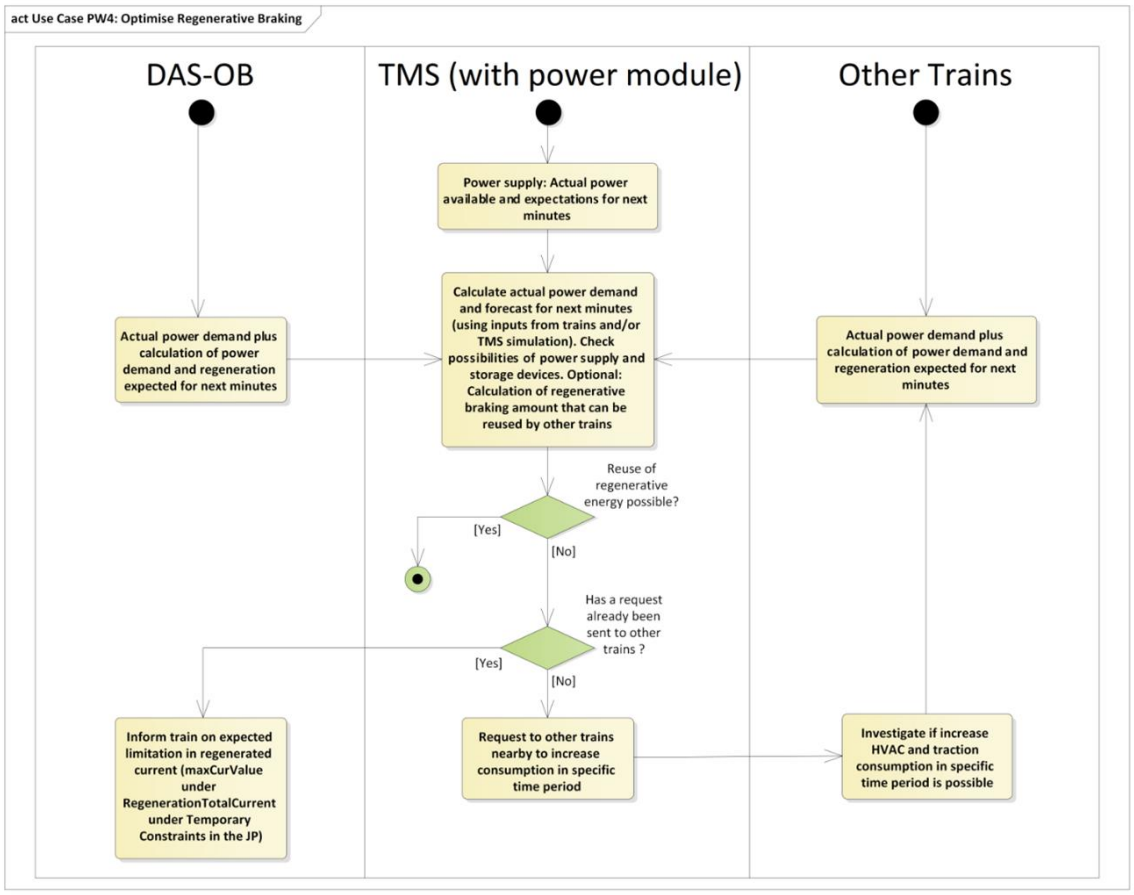


Figure 23: Activity Diagram – PW4

In this use case, the train communicates its expected power consumption for the next 5 to 15 minutes. The TMS collects the expected consumption and regeneration from all trains. When other trains nearby are able to use all regenerated energy, no further action is needed. In addition, if it is possible to store the surplus of regenerated energy in a trackside energy storage device, no further action is needed. If this is unlikely, the TMS will request trains nearby to increase HVAC and traction consumption in a specific time period. The DAS-OB on these trains investigates if the request can result in an increased consumption and give feedback to the TMS. When other trains cannot consume the expected regeneration, TMS will inform the DAS-OB that it will not be possible to inject the expected current.

It is clear that this use case can only function when the quality of calculation of expected power demand and regeneration for the next minutes is sufficiently accurate and the TMS is able to combine all this information in near real time.

Optimisation of regenerative braking is highly dependent on the behaviour of drivers, other consumers and also the supply system. This use case is likely only to be applicable to DC traction systems with irreversible power supply. In the case of AC traction systems, this braking energy is naturally sent back to the network.

In the PW3 and PW4 Use Cases, the DAS needs to be able to calculate the expected energy data for the next minutes and transmit this via the status message to DAS-TS. It is worth noting that self-propelled trains do not need to send such messages, because the TMS is already aware that they have no power consumption. However, for hybrid trains it is still useful to send the expected energy data to the TMS even when they are in autonomous mode.

This use case is described in the activity diagram shown in Figure 23 and in the sequence diagrams shown in Figure 24, Figure 25 and Figure 26. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	BoardAdviceCalculation(DAS-O)

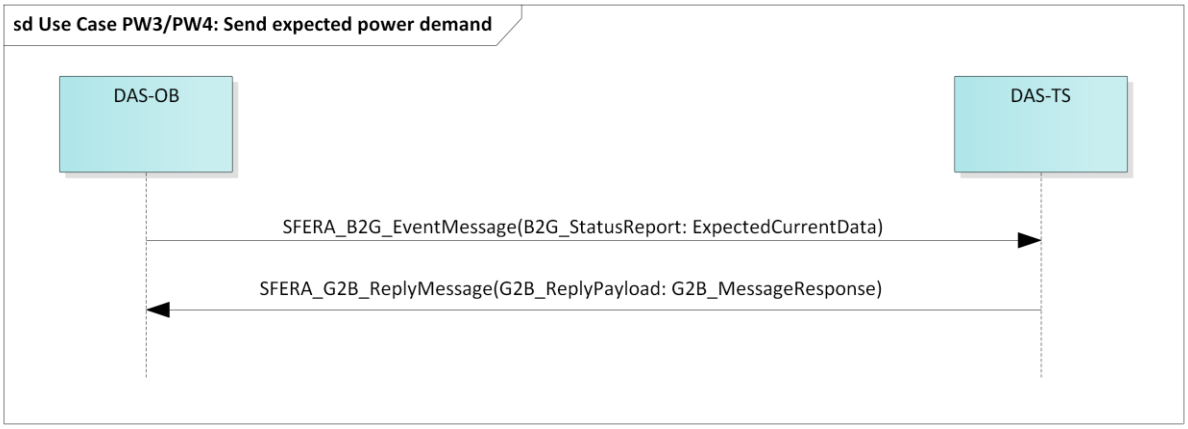


Figure 24: Sequence Diagram – PW3 and PW4 (Train sends expectedCurrentData)

PW4 can result in a request to some DAS-OB to maximise their power offtake. PW2 and PW3 can result in cases where a quick response is crucial. In order to avoid excessively long calculation and processing times, DAS-TS can also request some DAS-OB to minimise the power offtake. The DAS-TS can end the advice message with a 'no advice' indication.

The advice messages from DAS-TS regarding power offtake and temporary constraints on total current (traction or regeneration) will be included in the JP as Temporary Constraints, as illustrated in Figure 25 and Figure 26.

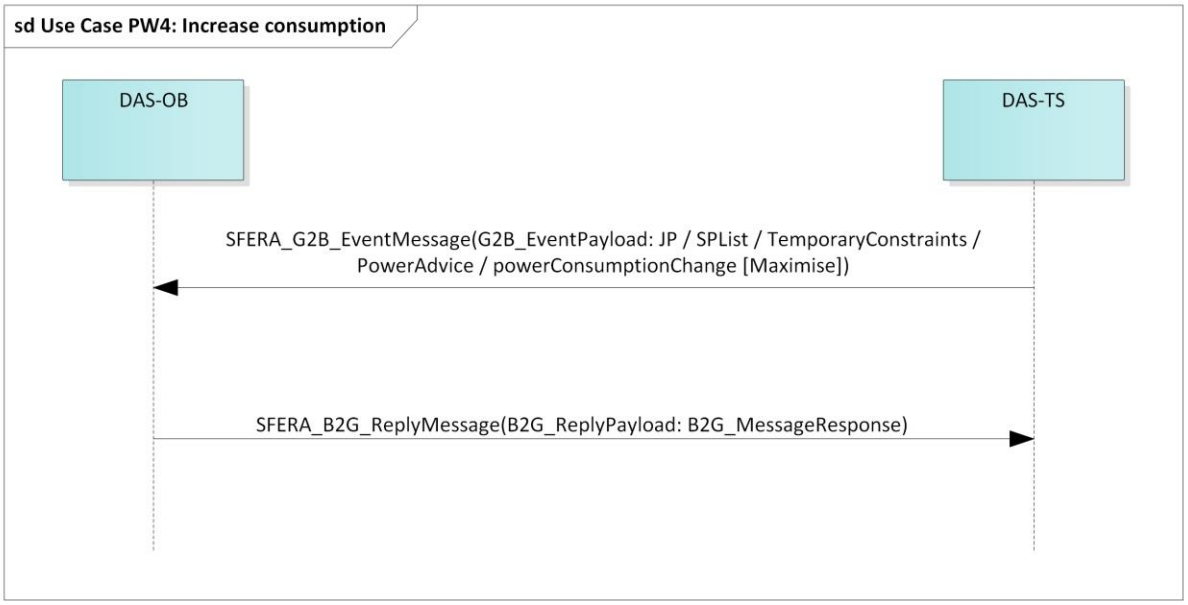


Figure 25: Sequence Diagram – PW4 (Increase consumption)

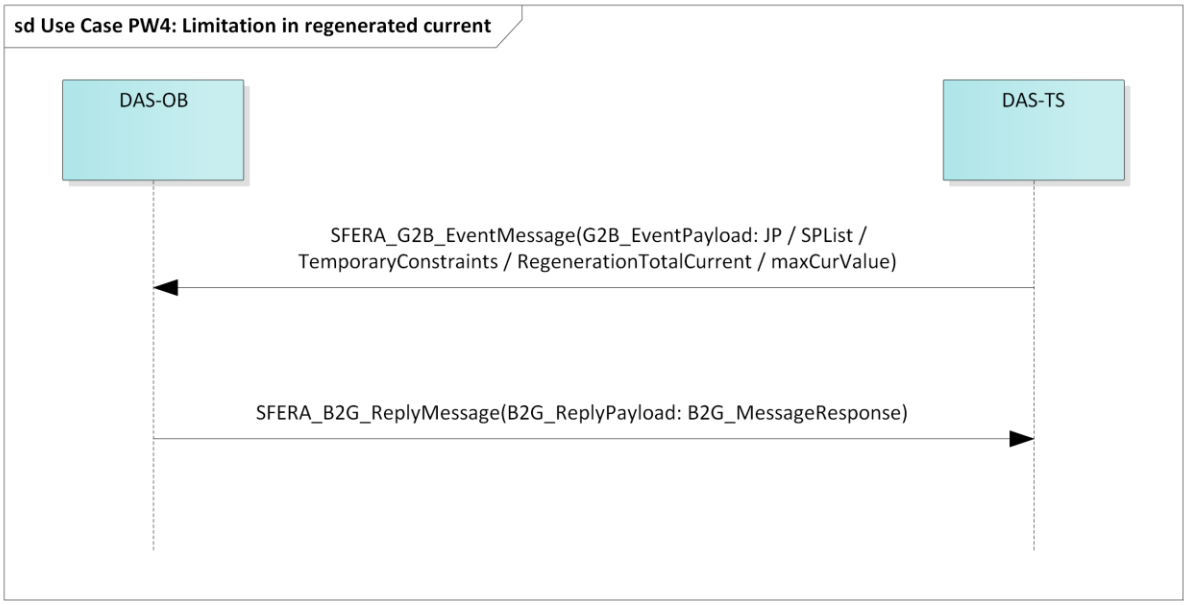


Figure 26: Sequence Diagram – PW4 (Limitation in regenerated current)

5 SR: Status Report

5.1 SR1: Position and Speed Information

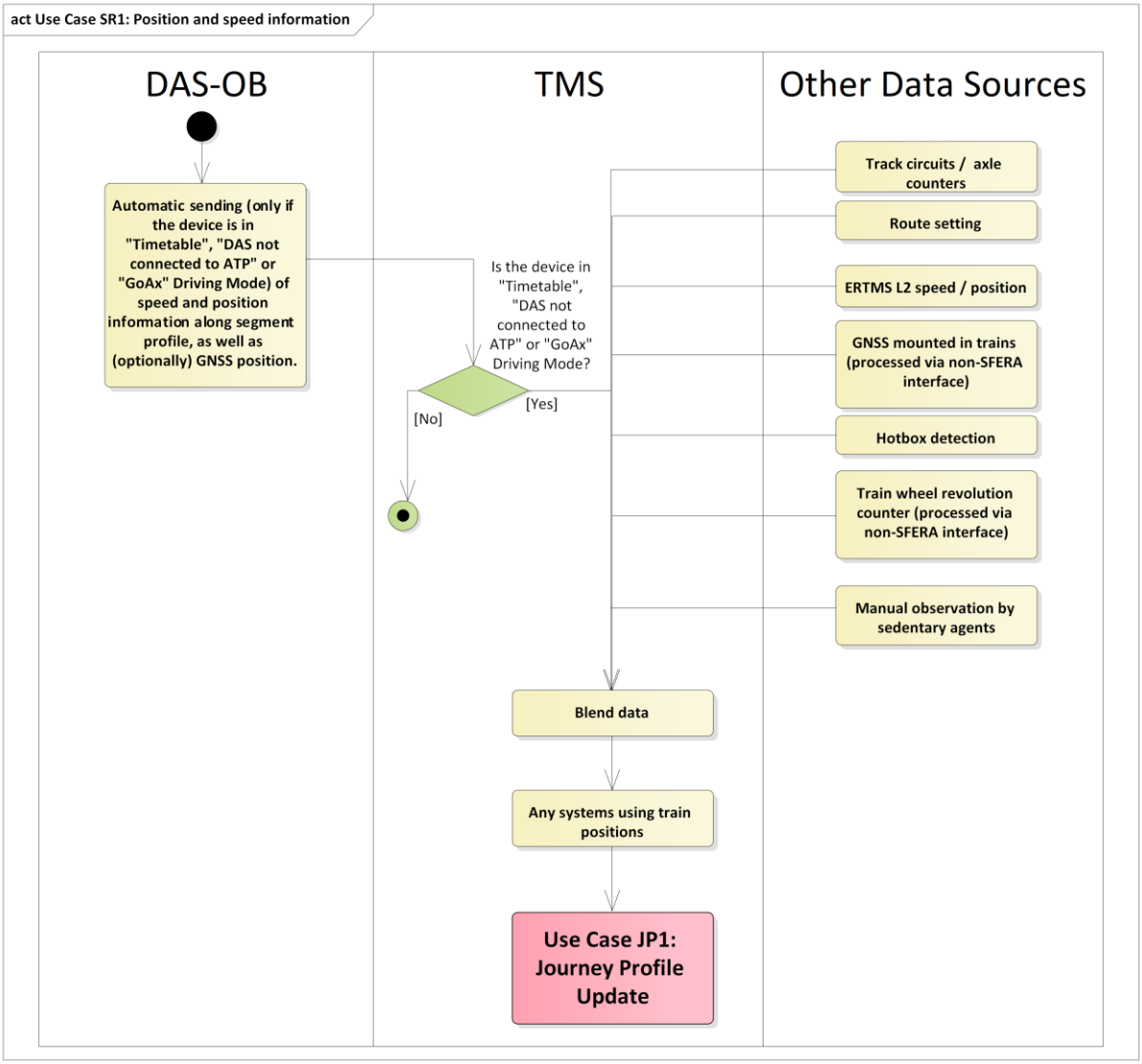


Figure 27: Activity Diagram – SR1

In this use case, the DAS device sends speed and position information to the TMS, which combines it with other data sources to elaborate train position updates that are as frequent and precise as possible. If the TMS detects a deviation between the planned JP and the position of the train, the TMS will re-evaluate the JP via Use Case JP1.

In the “inner/outer control loop” concept presented in Chapter 1 of the IRS, the TMS usually uses existing train localisation systems in the feedback loop. In cases where these systems do not give very frequent information, receiving data directly from the DAS will permit making the outer loop more reactive.

The DAS-TS will only take into account the Status Report coming from devices that have “Timetable”, “DAS not connected to ATP”, “GoA1”, “GoA2”, “GoA3” or “GoA4” driving modes. The DAS-OB shall also restrict the sending of status reports if it is in “Inactive” or “Readonly” mode.

The option of sending the GNSS position has been provided for multiple reasons:

- Sending an item of unaltered source information can be used to validate the position of the virtual balises;
- Sending unaltered source data may help in finding the root cause in case of imprecise DAS advice messages;
- To send the position in “SUBSET-126” format, the handheld device will make a projection of the measured GNSS position on the SP referenced by the last JP received. If the train was rerouted without being able to obtain an update of the JP (e.g. loss of connection or late update by the TMS), this position on the SP cannot be directly considered by the TMS. To have a reliable location in this situation, it is better for the IM to have the unaltered source information to remake the projection considering the real route taken by the train.

This use case is described in the activity diagram shown in Figure 27 and in the sequence diagram shown in Figure 28. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4
DAS Connectivity	Connected
DAS Architecture	All

This use case can be triggered on a cyclical basis.



Figure 28: Sequence Diagram – SR1

5.2 SR2: Train Estimates Late Arrival at Future Timing Point

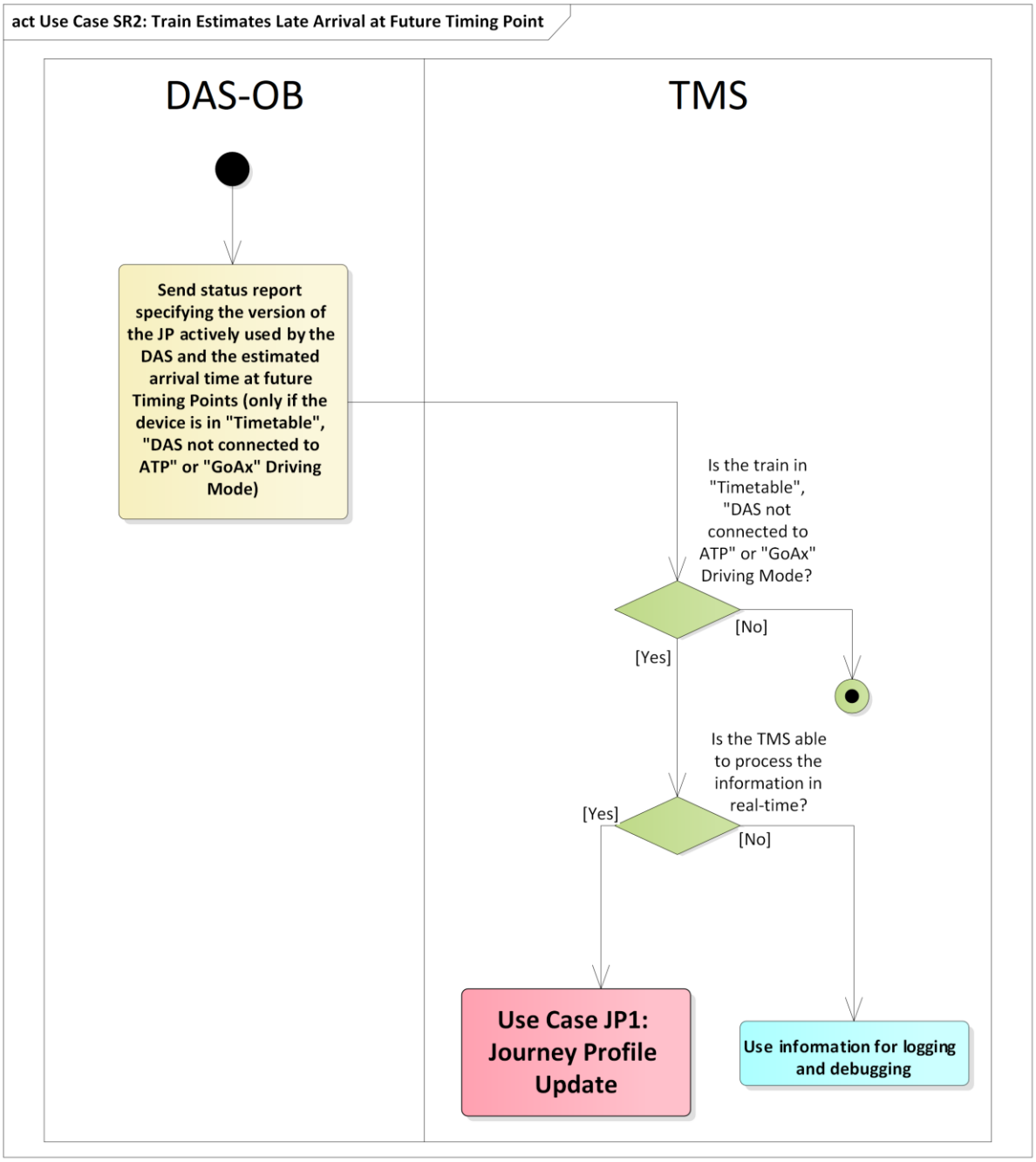


Figure 29: Activity Diagram – SR2

If an on-board DAS realises that the train cannot respect one or more Timing Points given in the JP, it will communicate this information to the TMS on the ground. Depending on the functionalities implemented in the TMS, this information will either be used to initiate the calculation of a new JP, or logged for debugging in post-operational analysis. This use case will mainly arise in two situations:

- Information known by the DAS but not yet known or taken into account by the TMS (location, Train Characteristics change, etc.),
- Different results between the algorithms of the TMS and the DAS (this can be due to the algorithm itself or to differences in data used by the TMS and DAS).

In the “inner/outer control loop” concept presented in Chapter 1 of the IRS, this use case introduces the possibility of feedback loop information that could be included in a TMS project.

This use case is described in the activity diagram shown in Figure 29 and in the sequence diagram shown in Figure 30. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4
DAS Connectivity	Connected
DAS Architecture	All



Figure 30: Sequence Diagram – SR2

The DAS-TS will only take into account the Status Reports coming from devices currently in one of the following driving modes: “Timetable”, “DAS not connected to ATP” or “ATO GoAx”.

This use case can be triggered on a cyclical basis.

5.3 SR3: Change in Adhesion Conditions

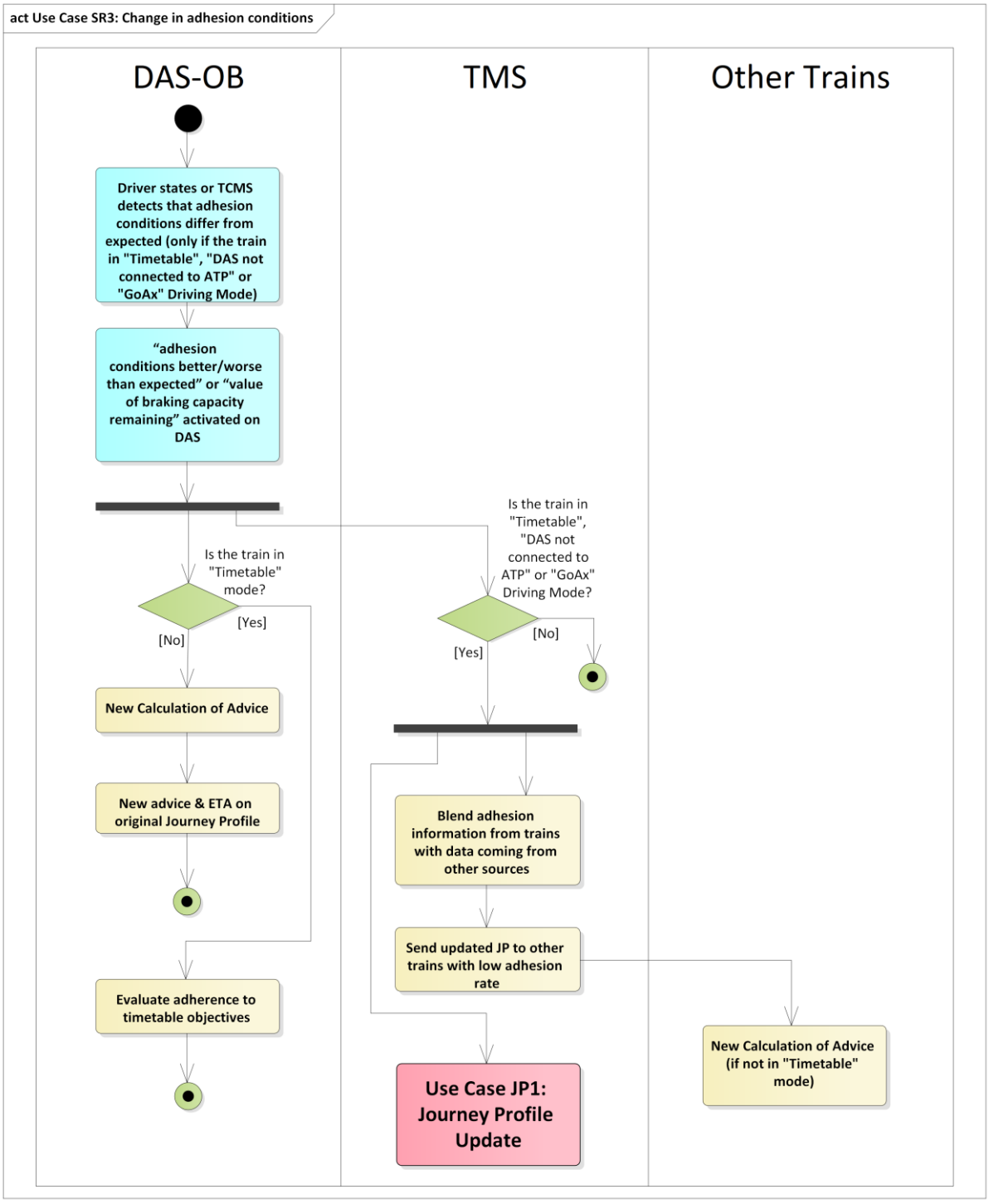


Figure 31: Activity Diagram – SR3

SFERA allows the driver to state (or the on-board DAS to detect) whether adhesion conditions are different from what was expected. This is very simply expressed by a message in which the on-board system communicates whether adhesion is "better" or "worse" than expected.

If the DAS-OB receives more accurate data (e.g. from the TCMS), the status message can include the percentage of remaining braking capacity.

In the “inner/outer control loop” concept presented in Chapter 1 of the IRS, this use case offers the possibility of introducing the adhesion conditions into the outer feedback loop.

The DAS-TS will only take into account the Status Report coming from devices that have “DAS not connected to ATP”, “GoA1”, “GoA2”, “GoA3” or “GoA4” driving modes. The DAS-OB shall also restrict the sending of status reports if it is in “Inactive” or “Readonly” mode.

The information is on the one hand used by the on-board DAS to perform calculations on its own course, and on the other hand it is transmitted to the TMS in order to share this fact with other trains and to be taken into account in the real-time traffic plan.

This use case is described in the activity diagram shown in Figure 31 and in the sequence diagrams shown in Figure 32 and Figure 33. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4
DAS Connectivity	Connected
DAS Architecture	All

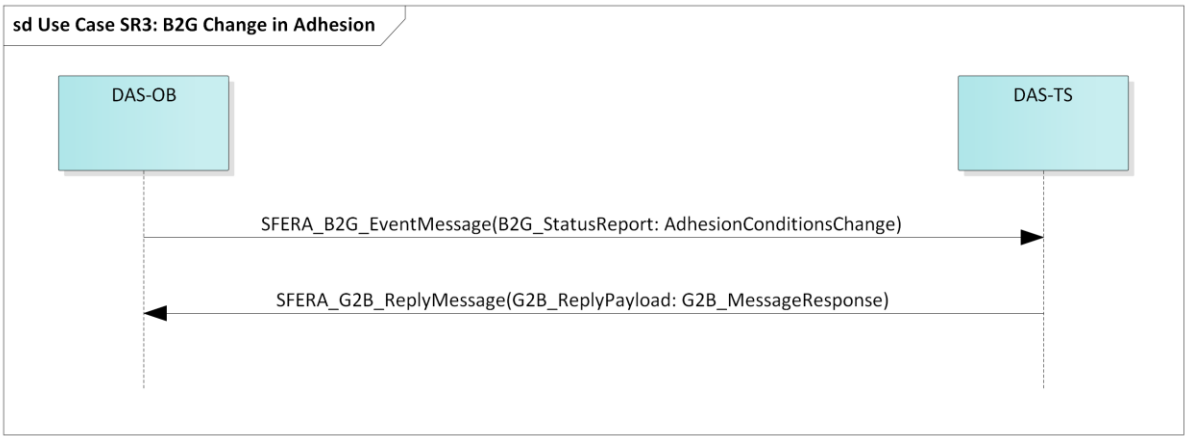


Figure 32: Sequence Diagram – SR3 (B2G Change in adhesion)

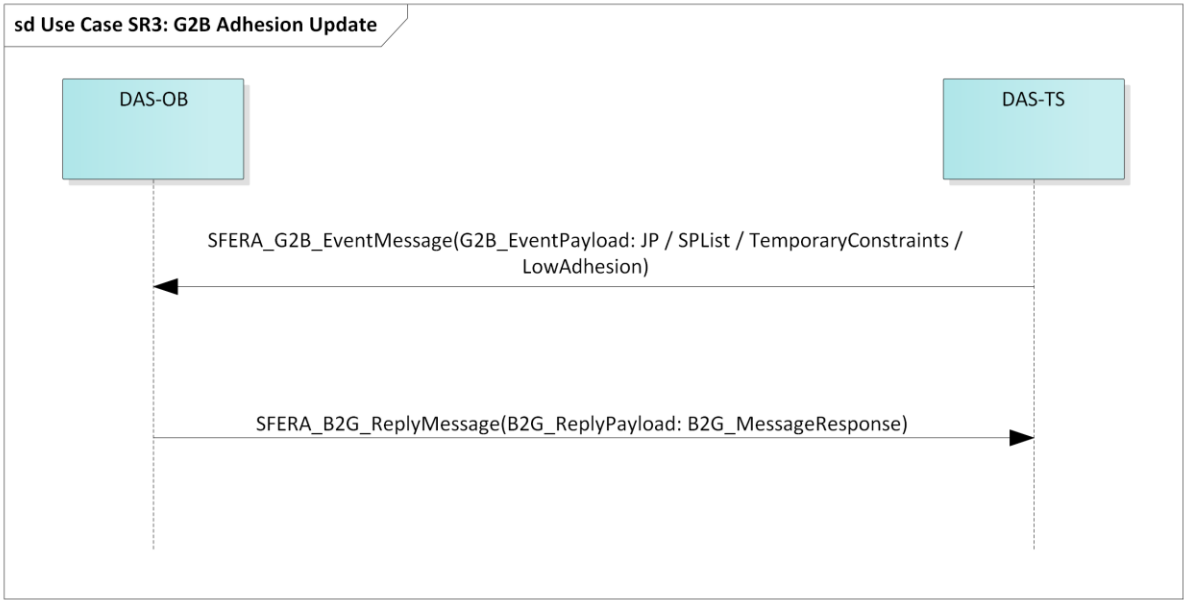


Figure 33: Sequence Diagram – SR3 (G2B Adhesion update)

6 DM: Driving Mode

6.1 DM1: Request Driving Mode of DAS Device

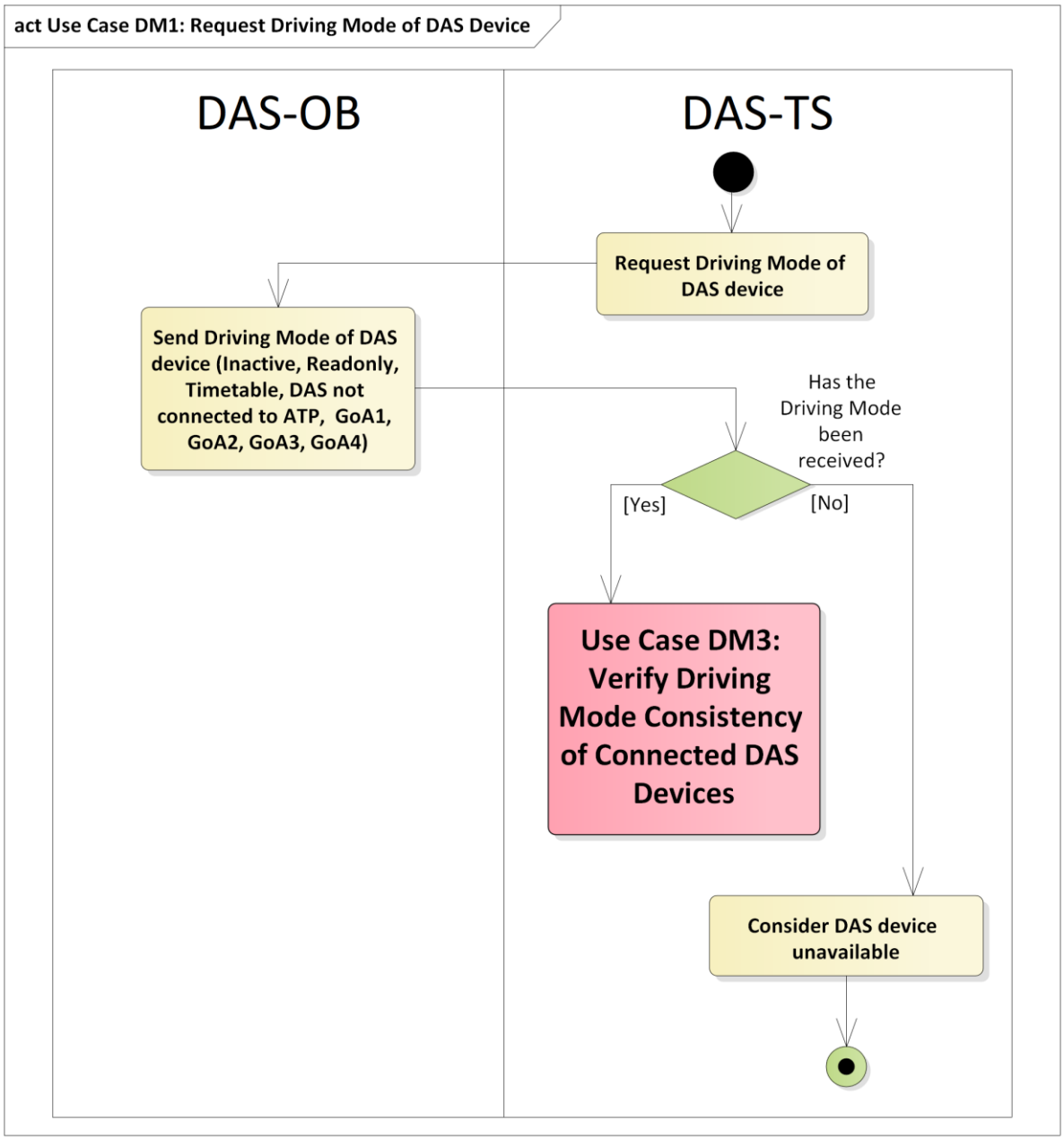


Figure 34: Activity Diagram – DM1

The DM1 Use Case is used by the TMS on the ground to request information on the Driving Mode (Inactive, Readonly, Timetable, DAS not connected to ATP, GoA1, GoA2, GoA3, GoA4) of the train system. This can be used as a keep-alive in case no communication occurs within a certain timespan, or if communication seems to be lost. If no reply comes from the on-board DAS, it is considered unavailable.

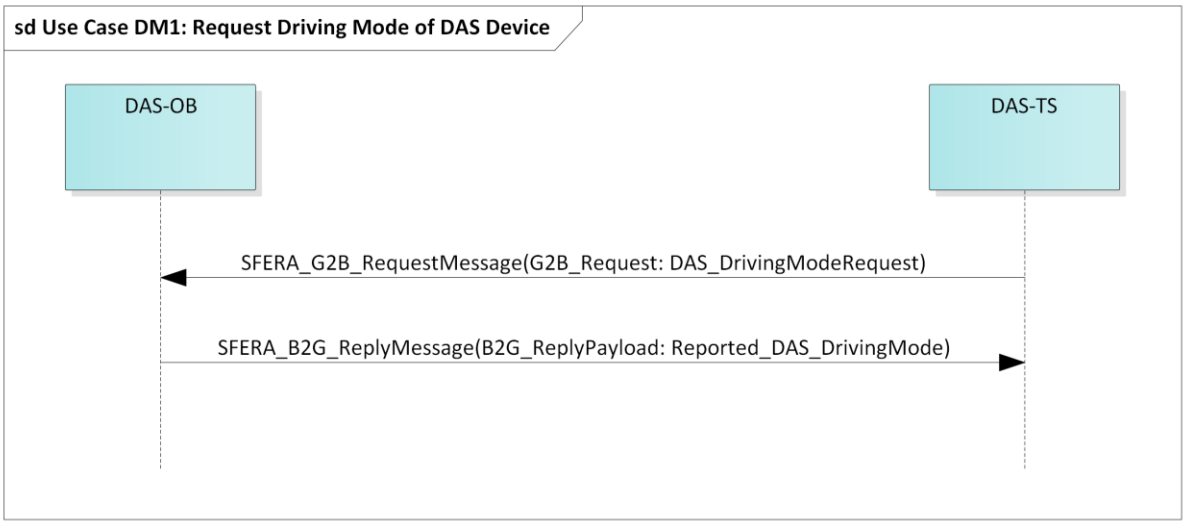


Figure 35: Sequence Diagram – DM1

This use case is described in the activity diagram shown in Figure 34 and in the sequence diagram shown in Figure 35. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	All

6.2 DM2: Send Driving Mode of DAS Device

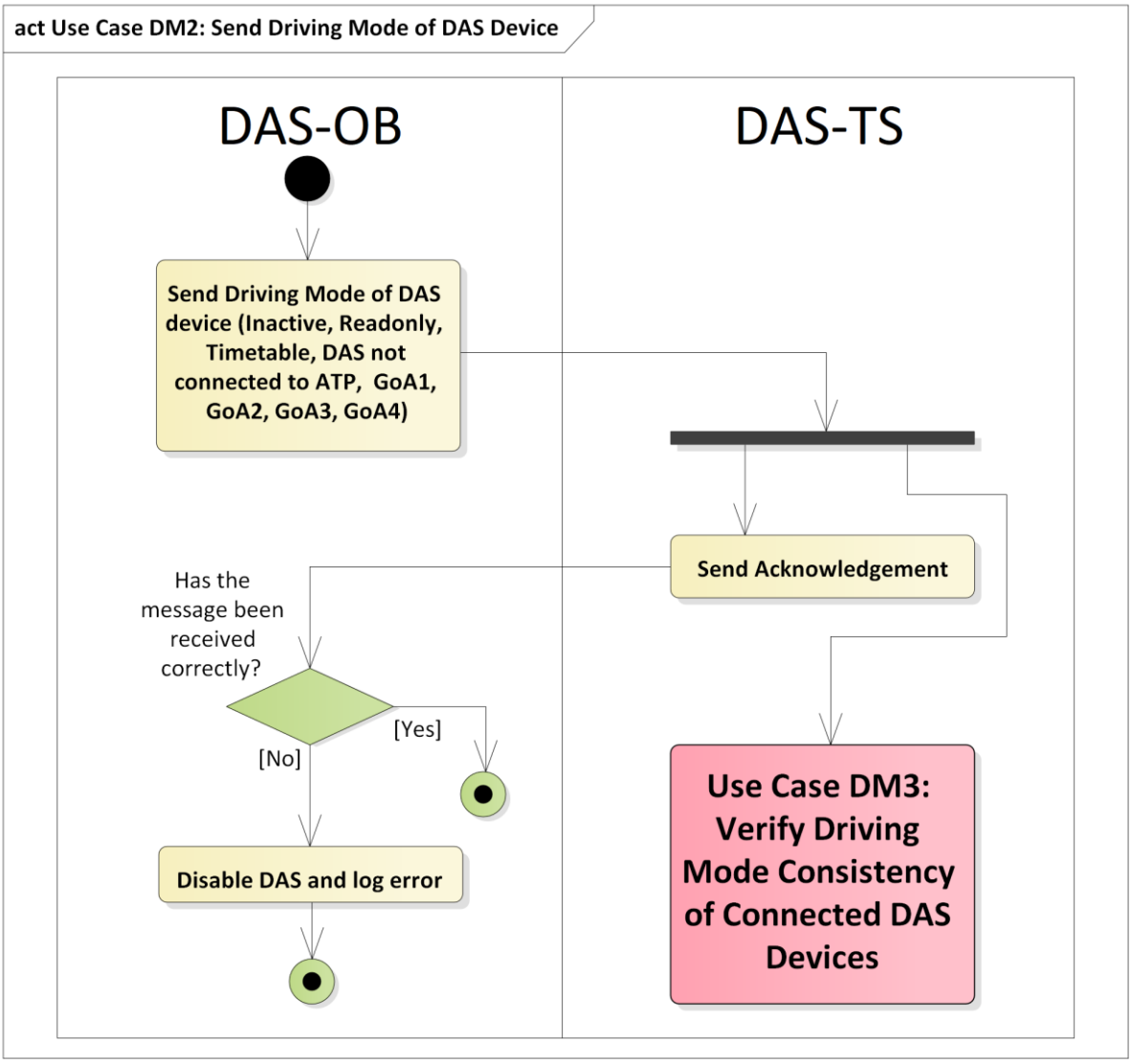


Figure 36: Activity Diagram – DM2

This case is used by the DAS-OB to communicate its Driving Mode to the ground. The driving mode can be:

- GoA4,
- GoA3,
- GoA2,
- GoA1,
- DAS not connected to ATP,
- Timetable,
- Readonly,
- Inactive.

This use case is described in the activity diagram shown in Figure 36 and in the sequence diagram shown in Figure 37. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	All

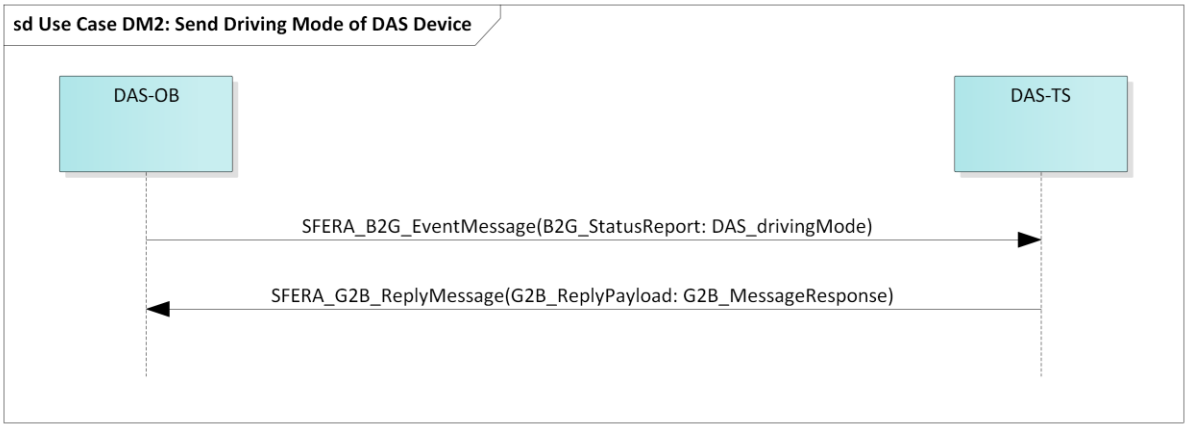


Figure 37: Sequence Diagram – DM2

6.3 DM3: Verify Driving Mode Consistency of Connected DAS Devices

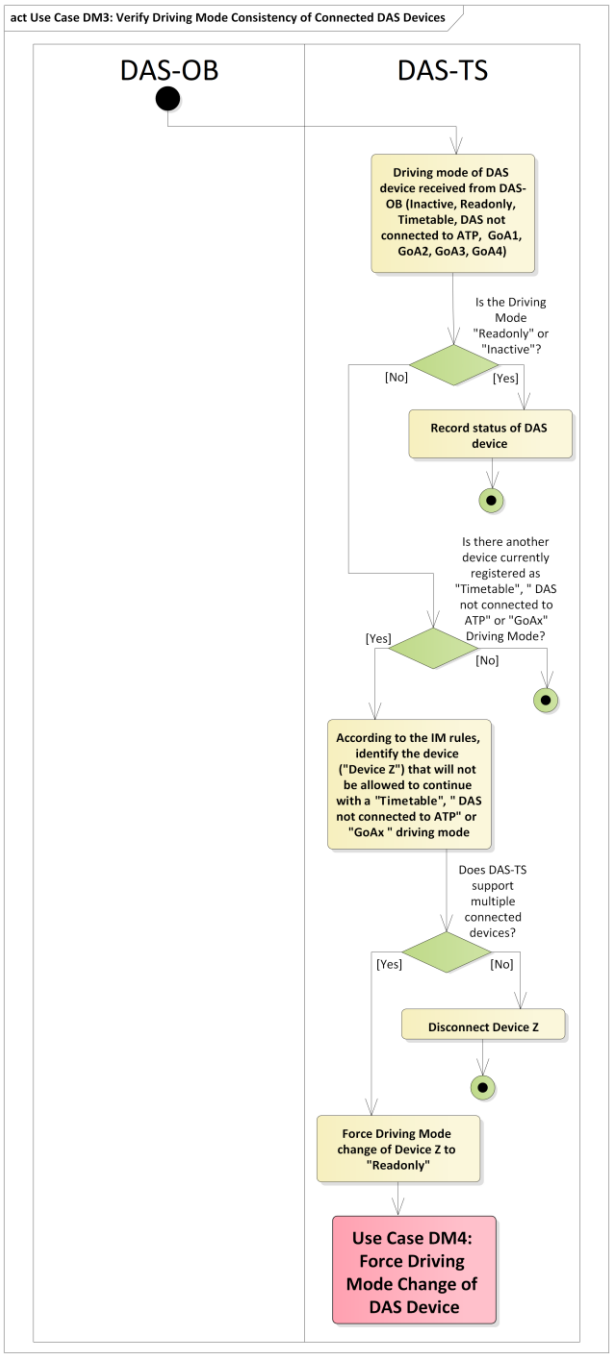


Figure 38: Activity Diagram – DM3

This use case is triggered when a driving mode change is received by the provider.

To be able to process the Status Reports, the DAS-TS shall implement the conditions to consider at most a single device as Timetable", "GoAx" or "DAS not connected to ATP" at any given time. So if a DAS-TS receives a "Timetable", "DAS not connected to ATP" or "GoAx" Driving Mode update, and another device already has one of these Driving Modes, the DAS-TS will have to make a decision on two aspects:

- Which DAS-OB is allowed to continue with this Driving Mode?
- Is the other DAS-OB allowed to stay connected with a "Readonly" status? If not, the DAS-TS triggers the disconnection of this device.

This use case is described in the activity diagram shown in Figure 38. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	All

6.4 DM4: Force Driving Mode Change of DAS Device

The DAS-TS may need to force a change of driving mode of a DAS-OB (typically to read-only, see DM3). In this case DM4 is used, which is a simple “message + acknowledgement” scheme.

This use case is described in the activity diagram shown in Figure 39 and in the sequence diagram shown in Figure 40. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	All

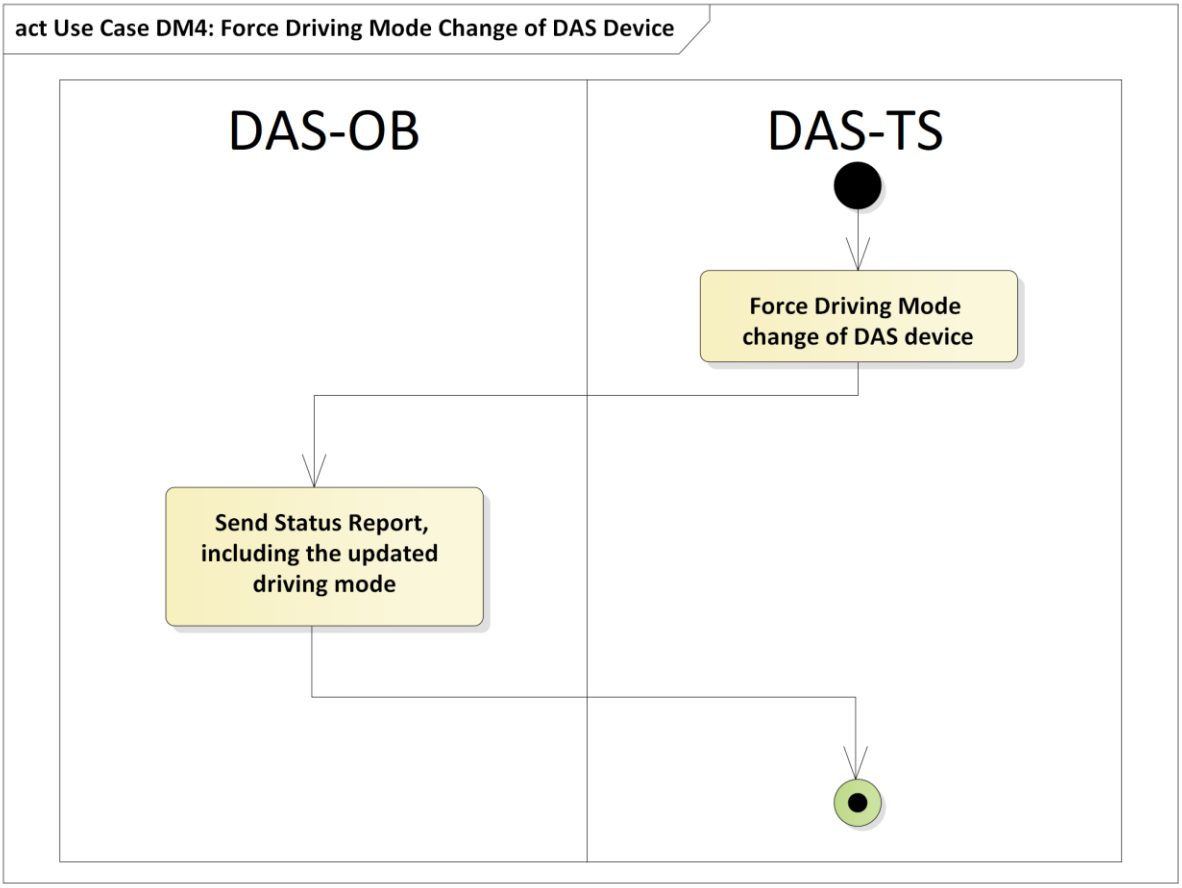


Figure 39: Activity Diagram – DM4

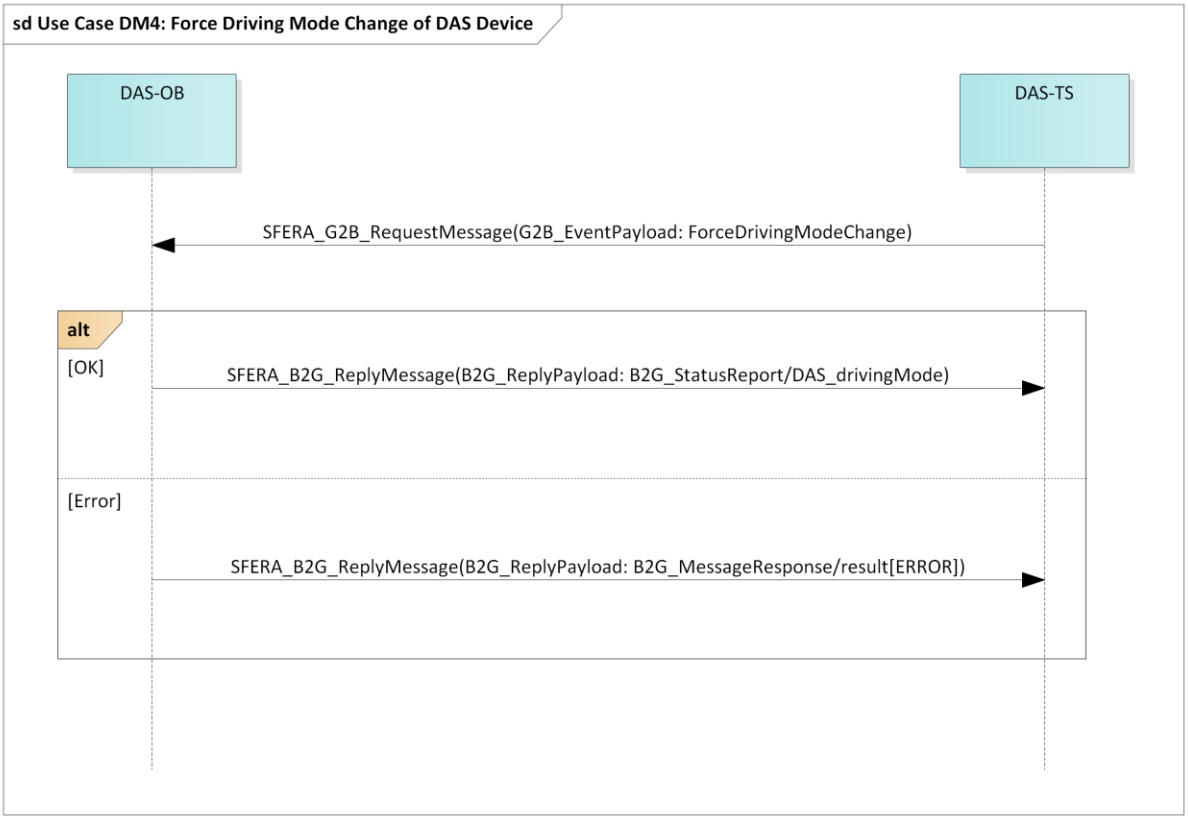


Figure 40: Sequence Diagram – DM4

7 CDC: C-DAS-C Advice

7.1 CDC1: C-DAS-C Advice Update

This is the main use case in C-DAS-C mode, describing how advice messages are sent to the DAS with an active connection to the DAS-TS.

Information generated internally (e.g. by dispatcher decisions), or externally from the trains or other sources, will be received by the TMS, which will use it to trigger advice updates. If the update impacts advice for one or more trains, the CDC3 Use Case is triggered for all of these trains.

This use case is described in the activity diagram shown in Figure 41. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	GroundAdviceCalculation(DAS-C)

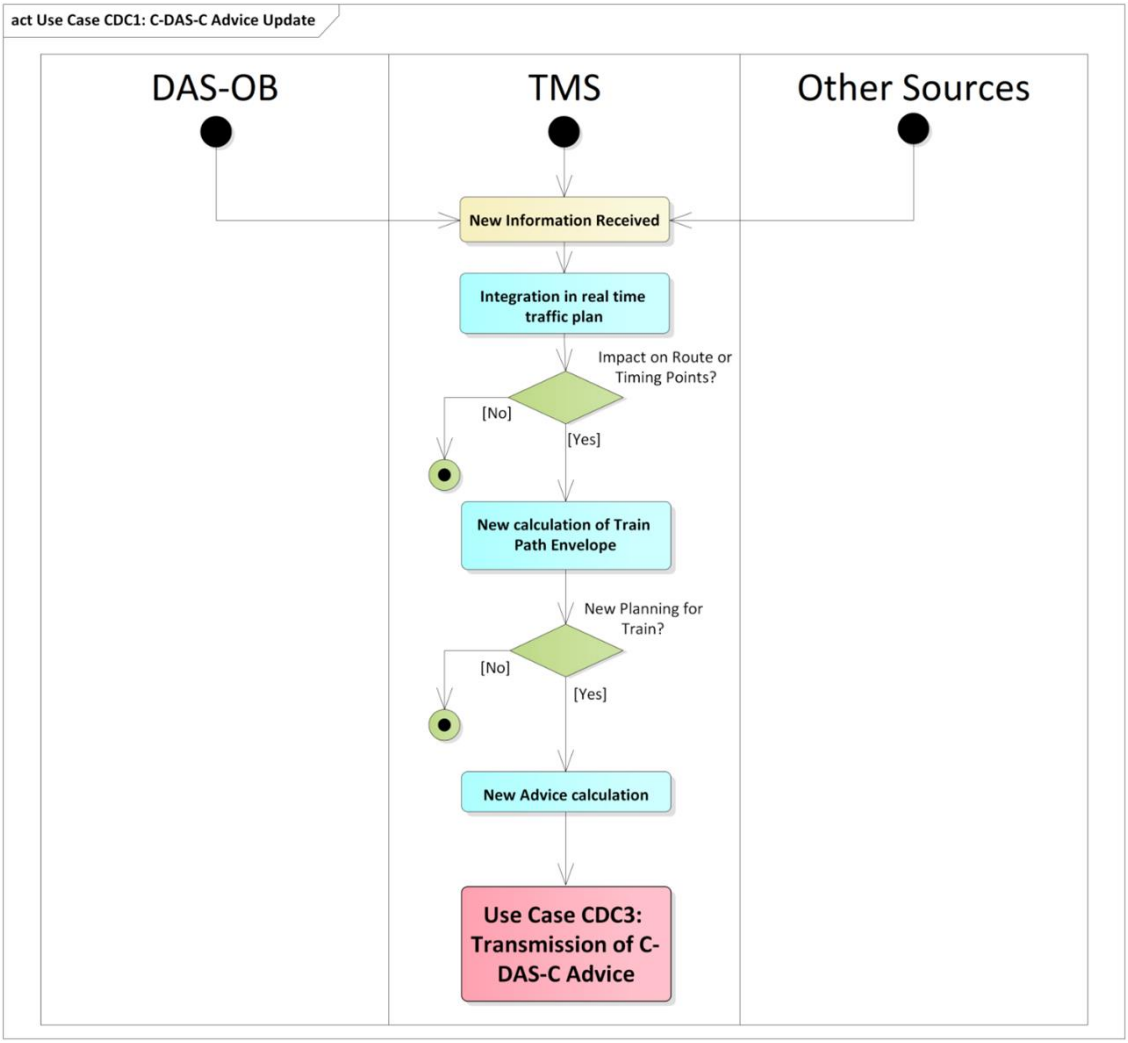


Figure 41: Activity Diagram – CDC1

7.2 CDC2: Request for C-DAS-C Advice

This use case will mainly be used by a C-DAS-C upon initialisation and in cases when the DAS needs to check for updates, for example when reconnecting after a journey portion where the on-board device has lost connection with the ground.

If the receiver requests C-DAS-C advice from the provider without specifying the version of the last known C-DAS-C advice message (e.g. upon first connection), one of three outcomes can be expected:

- if the data is available for the train, CDC3 is triggered;
- if the data is not available at this time, but will be available later, an error message is sent inviting the receiver to try again later;
- if the provider does not provide information for this train (now or later), an error message is sent, which terminates the connection.

If the receiver has provided the version of the C-DAS-C advice message in use, and the provider does not have a more recent version of the advice message available, an additional outcome is possible: simple acknowledgement.

This use case is described in the activity diagram shown in Figure 42 and in the sequence diagram shown in Figure 43. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	GroundAdviceCalculation(DAS-C)

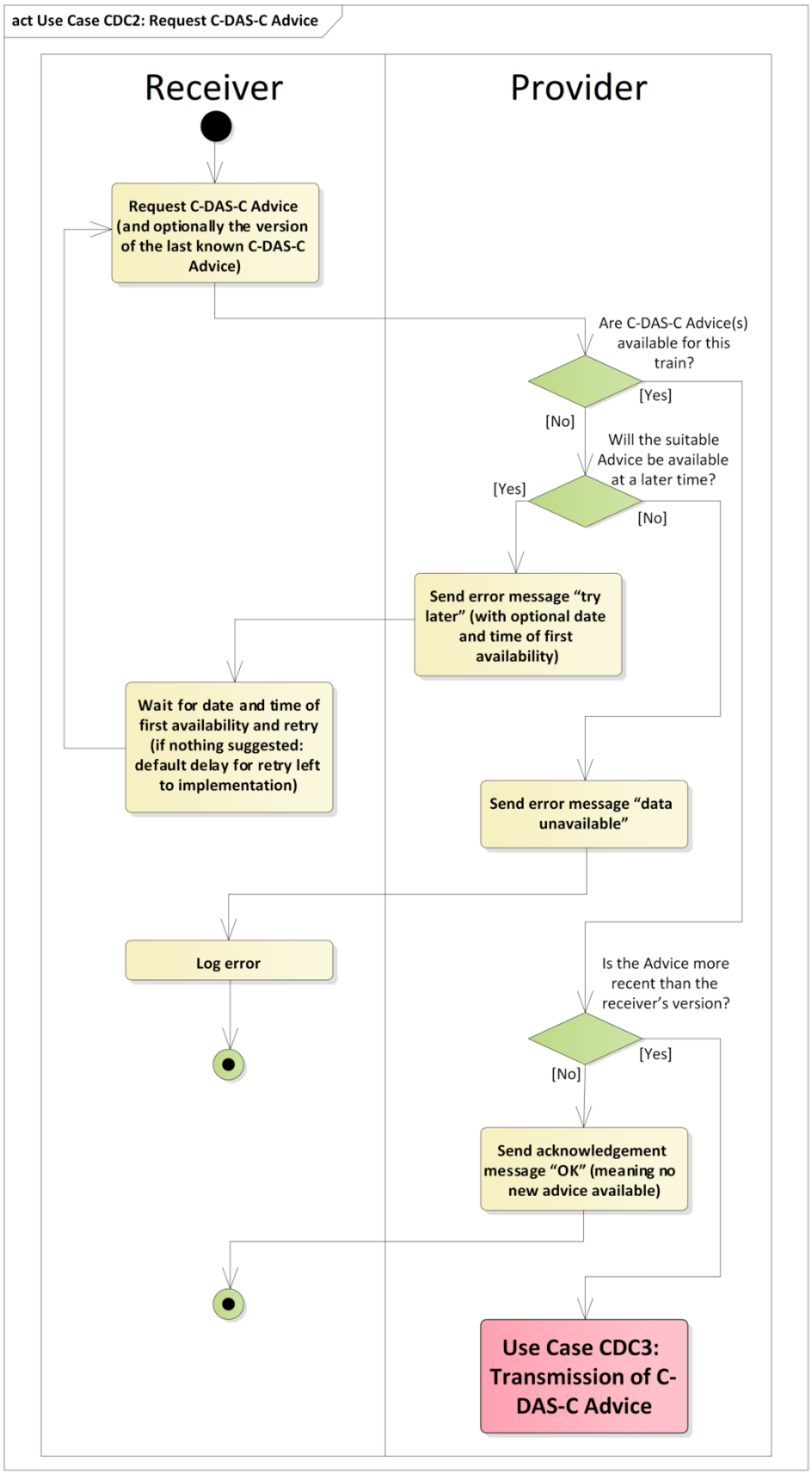


Figure 42: Activity Diagram – CDC2

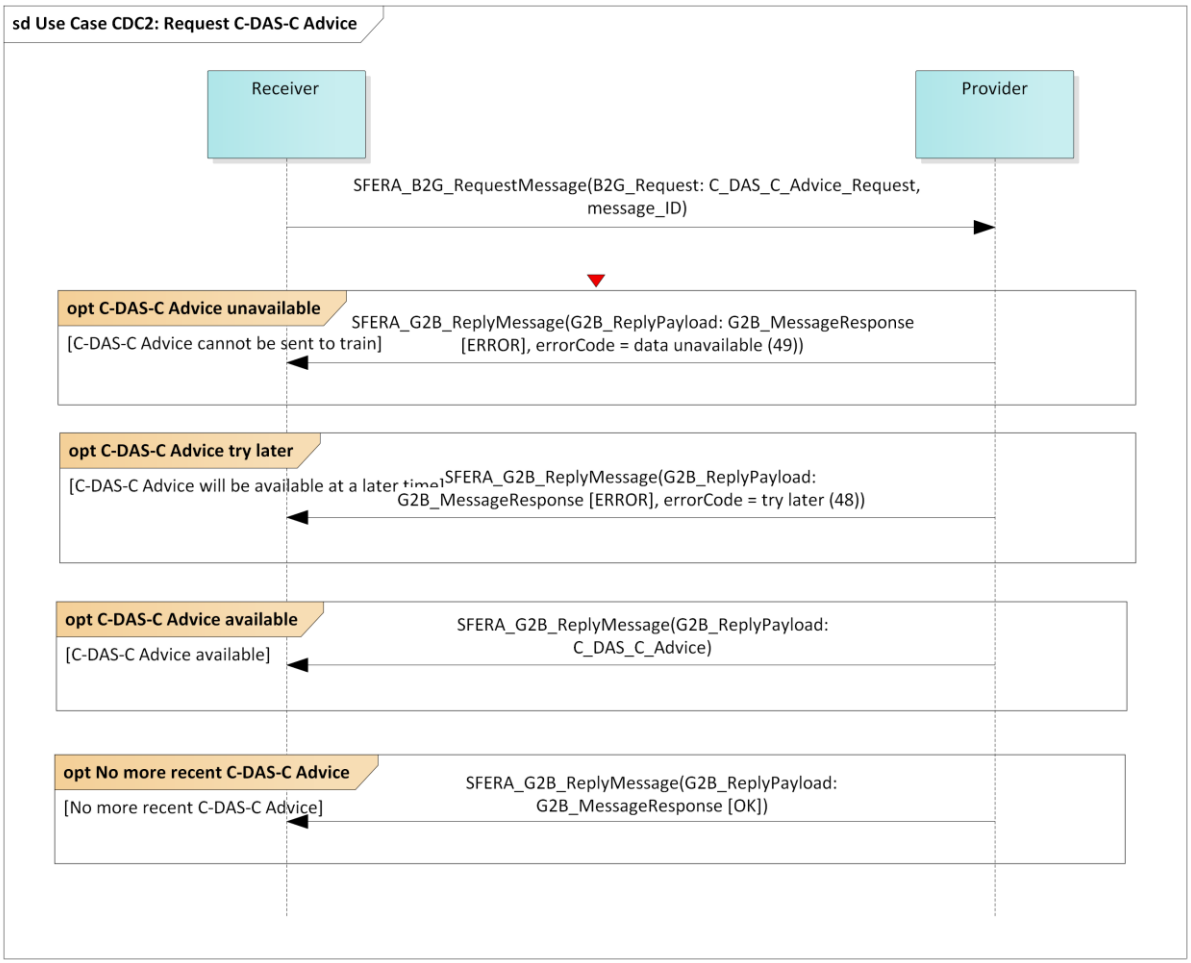


Figure 43: Transmission of C-DAS-C Advice and Acknowledgement

7.3 CDC3: Transmission of C-DAS-C Advice and Acknowledgement

CDC3 presents the sequence of actions used to ensure that the receiver gathers a complete set of information upon receiving a new C-DAS-C advice message. The C-DAS-C advice message can reference SPs not known by the receiver. If this is the case, Use Case JP4 will be used.

If the C-DAS-C advice message received could not be processed correctly by the receiver, an error message is sent to the provider.

The receiver confirms that the C-DAS-C advice message has been received in full (but not necessarily processed and actively used by the DAS-OB) by sending an acknowledgement message to the provider.

After receiving the updated C-DAS-C advice message, the on-board DAS verifies whether it can process it. If not, it has to send an error message to the provider. If the DAS-OB has been able to process the C-DAS-C advice message, a Status Report will be sent to the TMS specifying the new version of the advice message being used. Only after receiving this Status Report can the TMS consider that the train will take into account the new advice.

This use case is described in the activity diagram shown in Figure 44 and in the sequence diagrams shown in Figure 45 and Figure 46. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All

Operational Mode	Contexts in which this use case is applicable
DAS Connectivity	Connected
DAS Architecture	GroundAdviceCalculation(DAS-C)

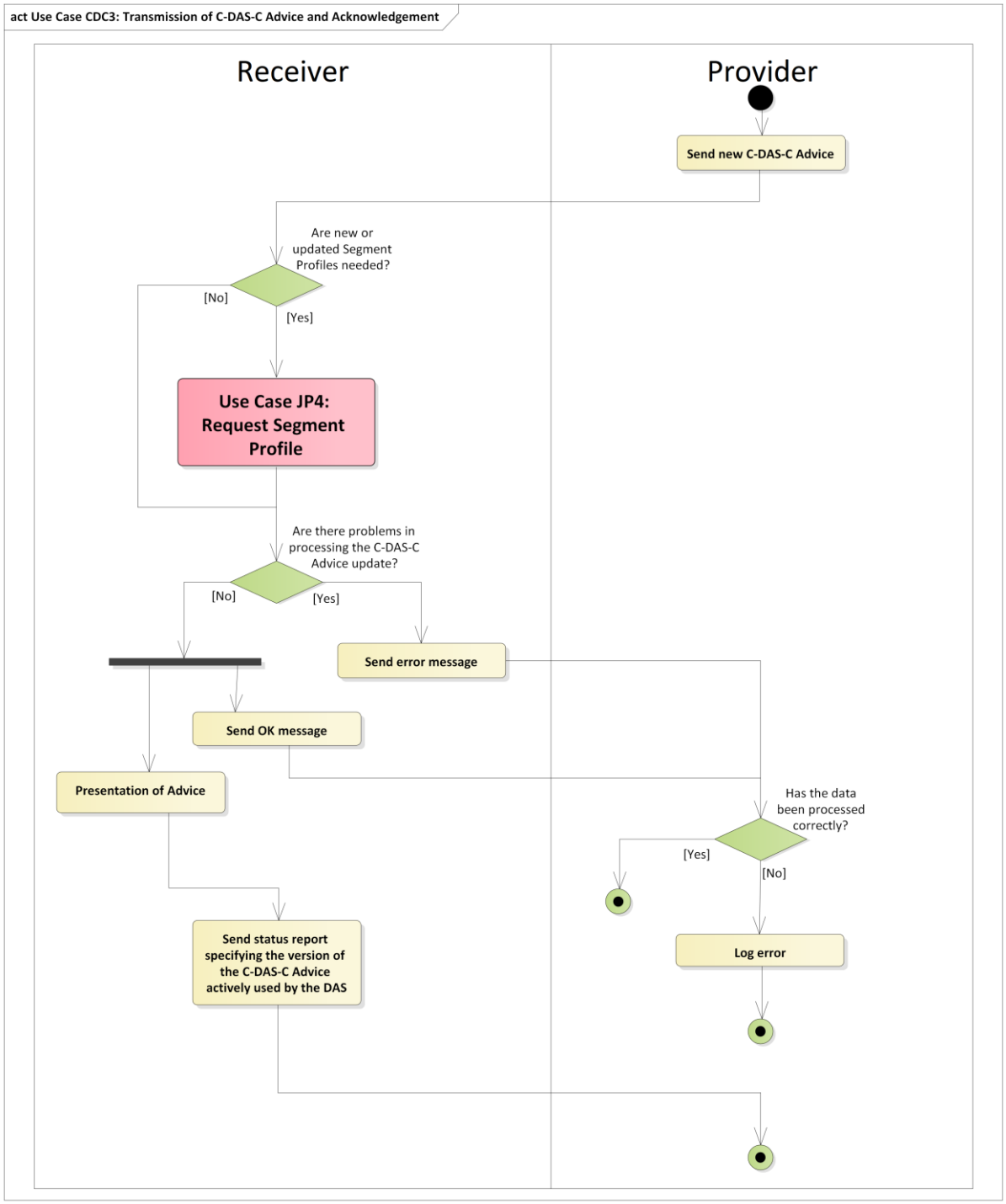


Figure 44: Activity Diagram – CDC3

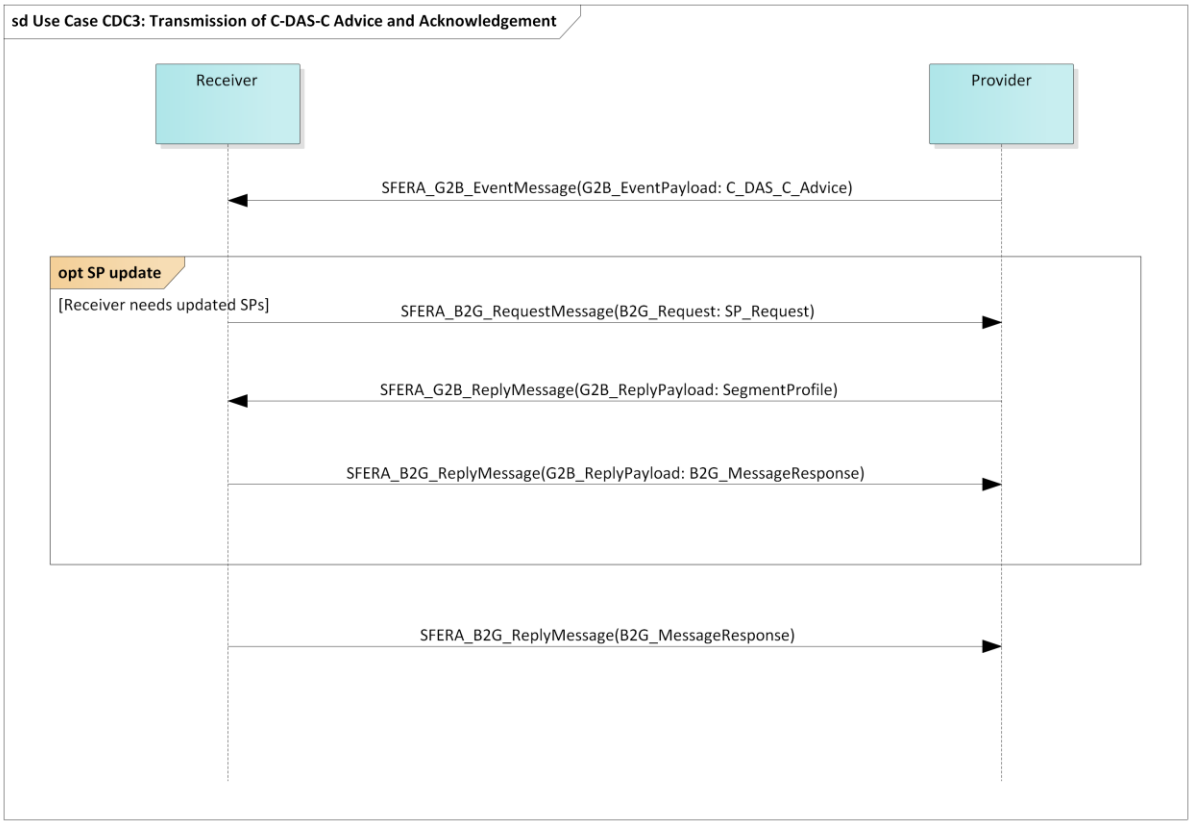


Figure 45: Sequence Diagram – CDC3 (C-DAS-C Advice and SP transmission)

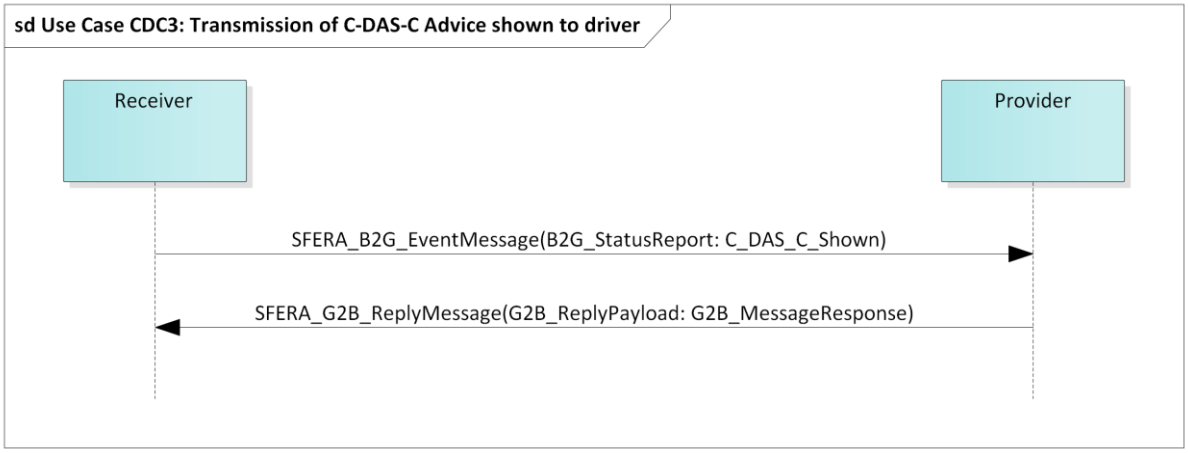


Figure 46: Sequence Diagram – CDC3 (Status Report)

8 AF: Additional Functions

8.1 AF1: Position and Speed from Ground (‘own train’ and related trains)

This use case has been implemented to cover different situations:

- the situations in which a DAS relying on GNSS positioning (e.g. a DAS on a tablet, therefore not having access to odometry) has a long portion of track without GNSS signals (e.g. underground inner-city lines, long tunnel, etc.);
- DAS implementations that do not have on-board positioning systems. In this cas, the DAS relies on a continuous speed/position feed from the DAS-TS (e.g. localisation through infrastructure sensors);

- DAS providing the “Related Train Information” function, which provides visibility on the trains sharing the same track, downline and upline.

Concerning the first situation, when the delay since the last known position passes a certain threshold, the DAS can request a position update from the ground for its last known position and speed coming from a source other than the DAS. Once the request has been fulfilled, the DAS-OB can calculate new advice for the rest of the journey. In the ‘inner/outer control loop’ concept presented in Chapter 1 of the IRS, the DAS uses an on-board system to implement the inner feedback loop concerning train position (odometry from ETCS, GNSS from the tablet or an on-board antenna, etc.). In some cases, this information can become unavailable (GNSS in a tunnel). This use case is an alternative plan to re-establish this inner feedback loop in such situations. It is only to be used when really necessary, as interrogating the ground for this information introduces latency and inaccuracy.

Concerning the second and third situation, such functionality has to be agreed on in the ‘related trains’ part of the handshake for the DAS-TS to autonomously send the position and speed from the ground.

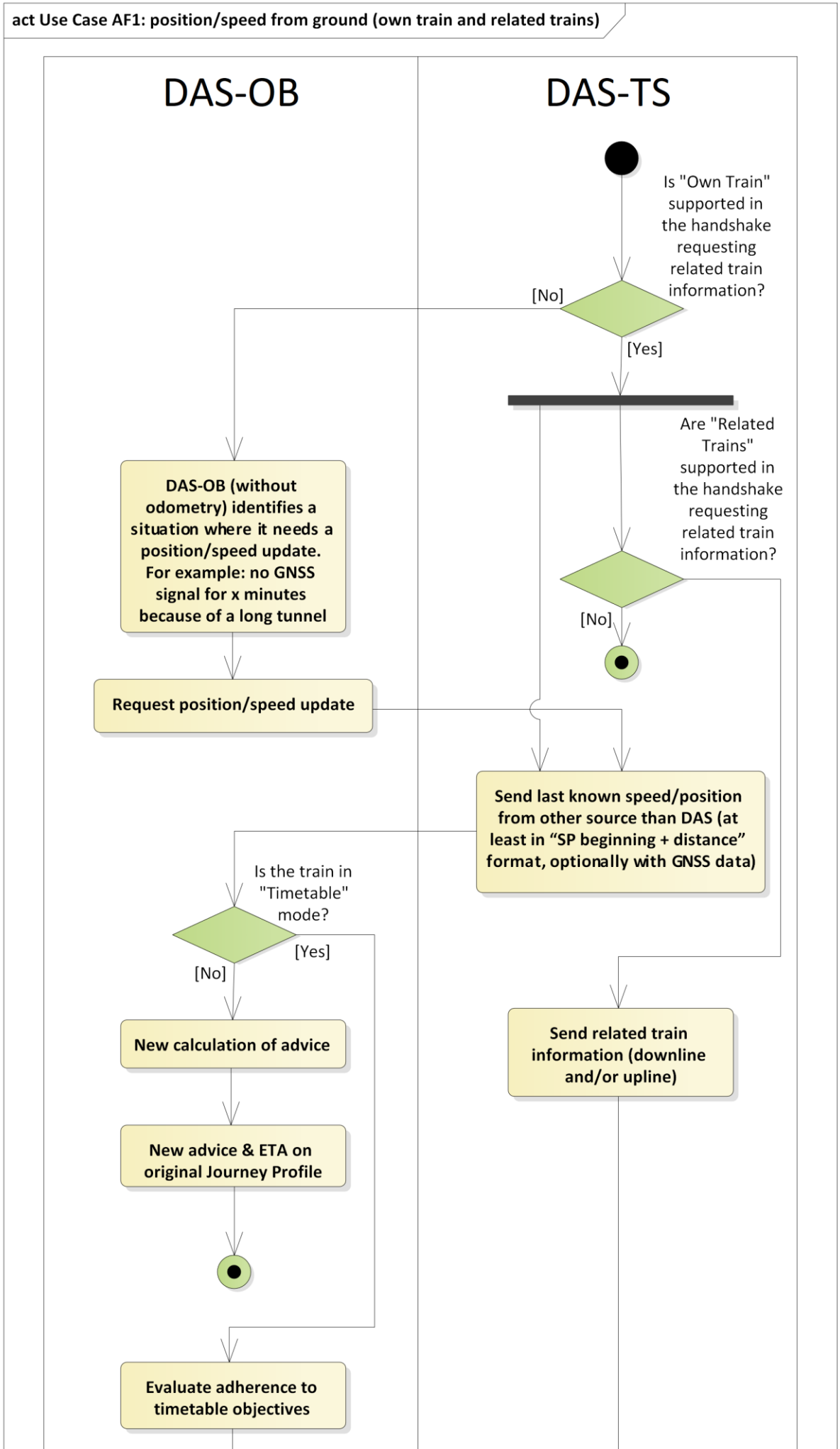

act Use Case AF1: position/speed from ground (own train and related trains)
DAS-OB
DAS-TS
Is "Own Train" supported in the handshake requesting related train information?
[No]
[Yes]
Are "Related Trains" supported in the handshake requesting related train information?
[No]
DAS-OB (without odometry) identifies a situation where it needs a position/speed update. For example: no GNSS signal for x minutes because of a long tunnel
Request position/speed update
Send last known speed/position from other source than DAS (at least in "SP beginning + distance" format, optionally with GNSS data)
Is the train in "Timetable" mode?
[Yes]
[No]
New calculation of advice
New advice & ETA on original Journey Profile
Send related train information (downline and/or upline)
Evaluate adherence to timetable objectives

Figure 47: Activity Diagram – AF1

This use case is described in the activity diagram shown in Figure 47 and in the sequence diagram shown in Figure 48. It is applicable in the following contexts:

Operational Mode	Contexts in which this use case is applicable
DAS Driving Mode	All
DAS Connectivity	Connected
DAS Architecture	All



Figure 48: Sequence Diagram – AF1

9 Application to IM-RU Setup

The sequence diagrams associated with the use cases in this chapter have been described with the IM-Train Setup in mind (messages are exchanged directly between the DAS and the IM, with a push strategy). For the IM-RU setup, the RU back office acts as a middleman. In this case, the RU will need to choose if it wants to implement a caching mechanism. The caching mechanism, if implemented, would have as its main objective a reduction in the amount of data exchanged multiple times by the IM and RU. This would essentially concern the SP and TC objects being used by multiple trains every day and would also represent a performance advantage. The JPs could also be cached in cases where there is a widespread practice of multiple devices accessing the information concerning a single train service. As an example, the following paragraphs illustrate how the sequence diagrams for Use Case JP3 would be adapted in the IM-RU setup, presenting a push strategy for two cases: with and without caching.

9.1 Sequence Diagrams for the RU-IM Setup without an RU Cache

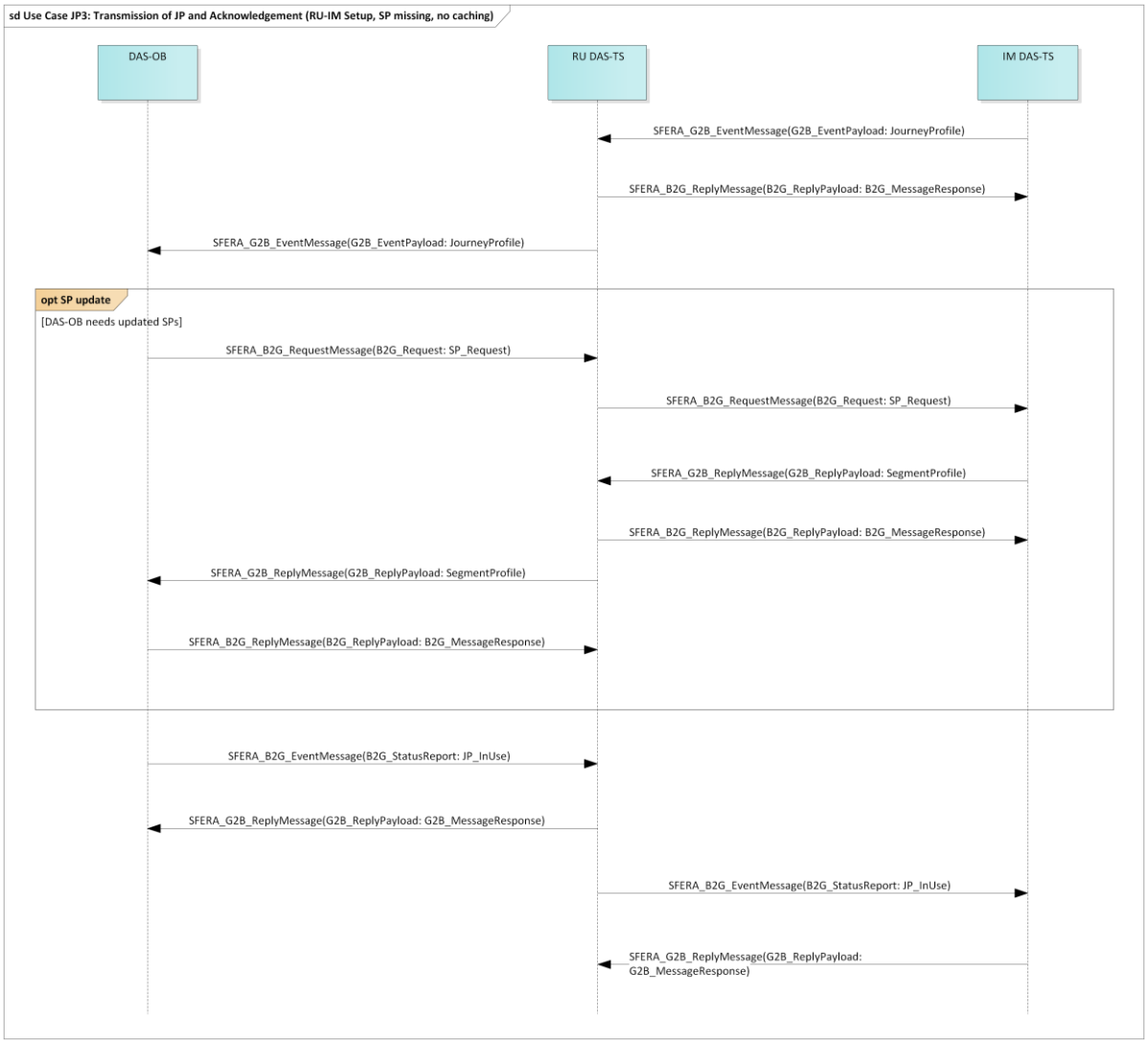


Figure 49: Sequence Diagram – RU-IM Setup example without an RU cache

The sequence diagram shown in Figure 49 presents an update of a JP. The RU DAS-TS receives the updated JP. It forwards this to the DAS-OB and in parallel acknowledges its receipt to the IM DAS-TS. This acknowledgement is only technical and shall not be considered as receipt by the DAS-OB. In the Segment Profile list of the JP, the DAS-OB finds some SPs that are not yet available on board and requests to receive them. The RU DAS-TS forwards the request to the IM DAS-TS. The IM DAS-TS replies with the intended SPs. RU DAS-TS acknowledges their receipt and in parallel forwards the SPs to the DAS-OB. The message response for receipt of the JP from the DAS-OB is given to the RU DAS-TS.

Once the DAS-OB considers it has all the information necessary to use the JP for calculations, a Status Report is sent to the RU DAS-TS, which is then relayed to the IM DAS-TS. From that moment on, the TMS can consider that the new JP will be taken into account by the driver or ATO.

9.2 Sequence Diagrams for the RU-IM Setup with an RU Cache

If the RU DAS-TS implements a caching mechanism, it will keep in memory the TCs and SPs it has received in the requests made by the different devices (see sequence diagram in

Figure 50). The IM and the RU could potentially agree on implementing a way of periodically sending in bulk the new versions of the SPs and TCs of the network.

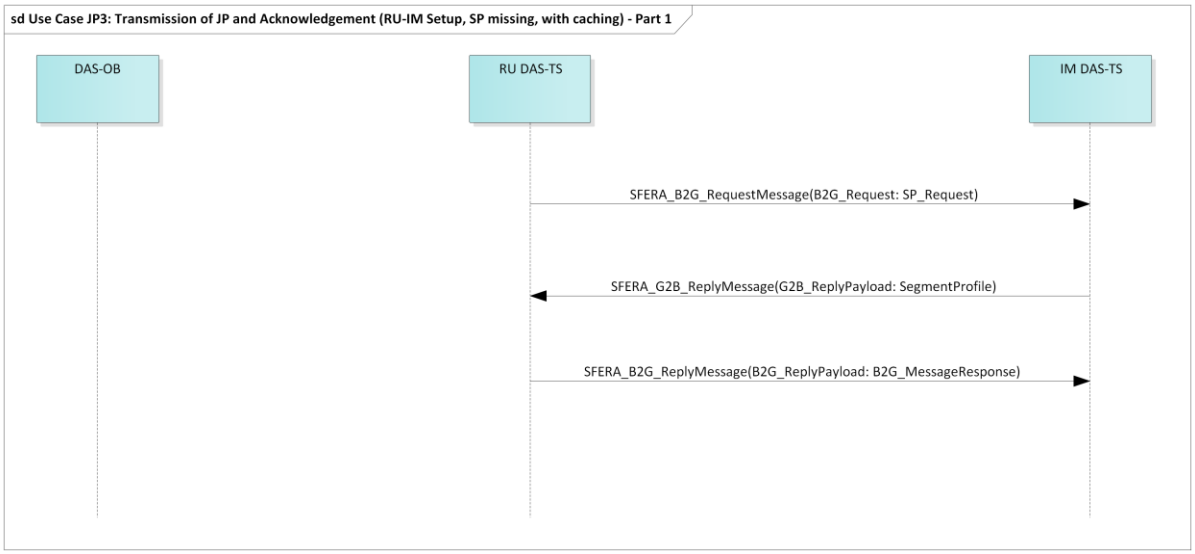


Figure 50: Sequence Diagram – Example of the RU backoffice caching a Segment Profile

When the IM DAS-TS transmits a JP update, the RU DAS-TS acknowledges receipt of the JP, and forwards the JP to the DAS-OB. The DAS-OB still needs to request the missing SPs. But as this data is cached in the RU DAS-TS, there is no need to forward the request to the IM DAS-TS. So the RU DAS-TS immediately replies with a message including the requested SPs. The DAS-OB replies with the message response on receipt of the SPs.

Once the DAS-OB considers it has all the information necessary to use the JP for calculations, a Status Report is sent to the RU DAS-TS, which is then relayed to the IM DAS-TS. From that moment on, the TMS can consider that the new JP will be taken into account by the driver or ATO.

This process is represented in the sequence diagram shown in Figure 51.

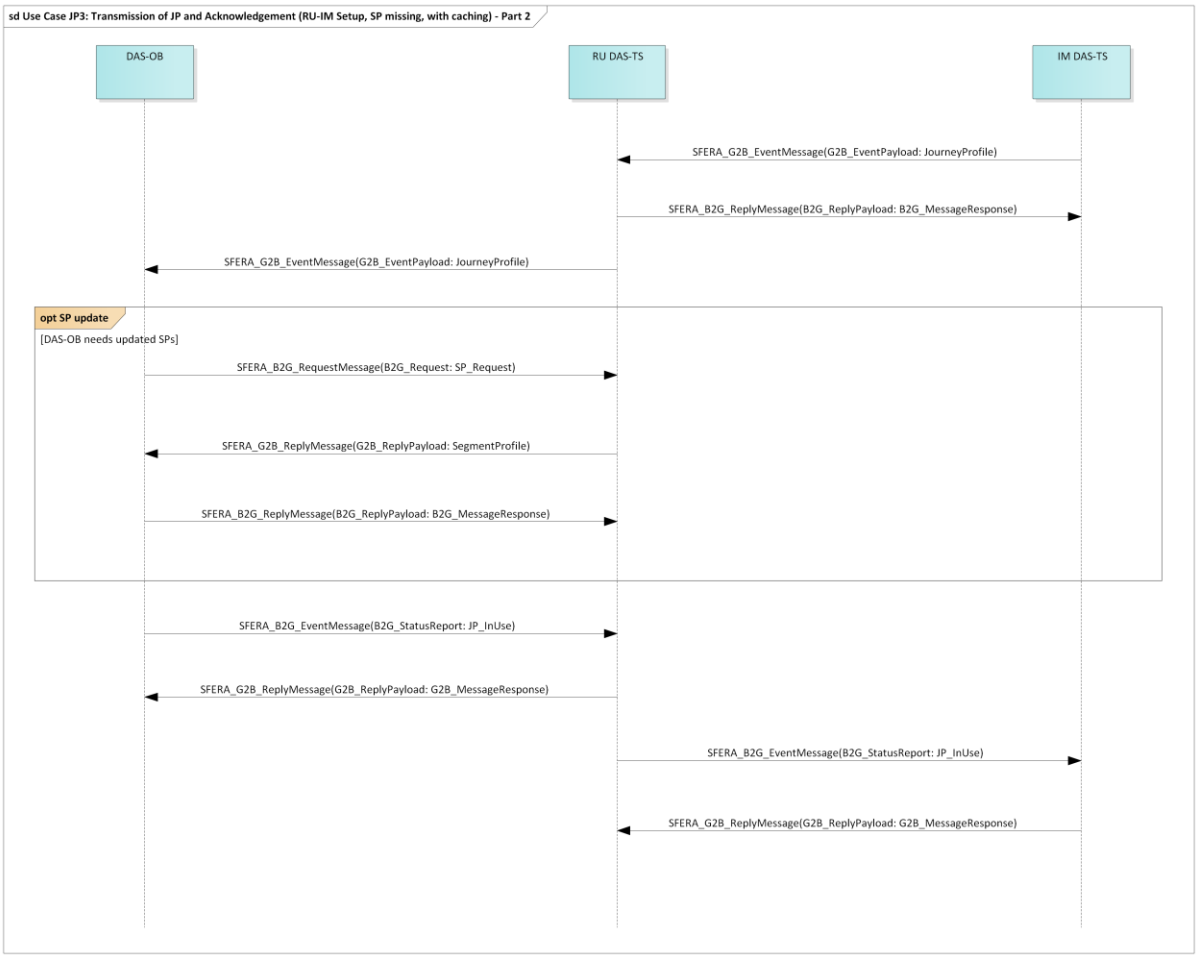


Figure 51: Sequence Diagram - IM-RU Setup example with RU cache

It is to be noted that the DAS-OB could detect errors in the SP that were not detected by the RU DAS-TS. In that case, the DAS-OB will send an error message to the RU DAS-TS, which will be relayed directly to the IM DAS-TS. This means that the IM DAS-TS could be subject to receiving error messages for SPs that it has sent multiple hours or days before.

9.3 Handshakes

In the IM-RU setup, each DAS-OB will implement connection and disconnection use cases with the RU DAS-TS for every train it needs information for. The RU DAS-TS will implement a level of abstraction, and implement DX1 with the IM DAS-TS from the moment at least one DAS-OB has successfully established the connection handshake with the RU DAS-TS, and implement DX2 from the moment all DAS-OB have successfully closed the connection, or have lost their connection for longer than the timeout taken into account by the RU DAS-TS. If a DAS-OB connects or disconnects to the RU DAS-TS when another DAS-OB is already in a connected state, no handshake will need to be performed between RU DAS-TS and IM DAS-TS.

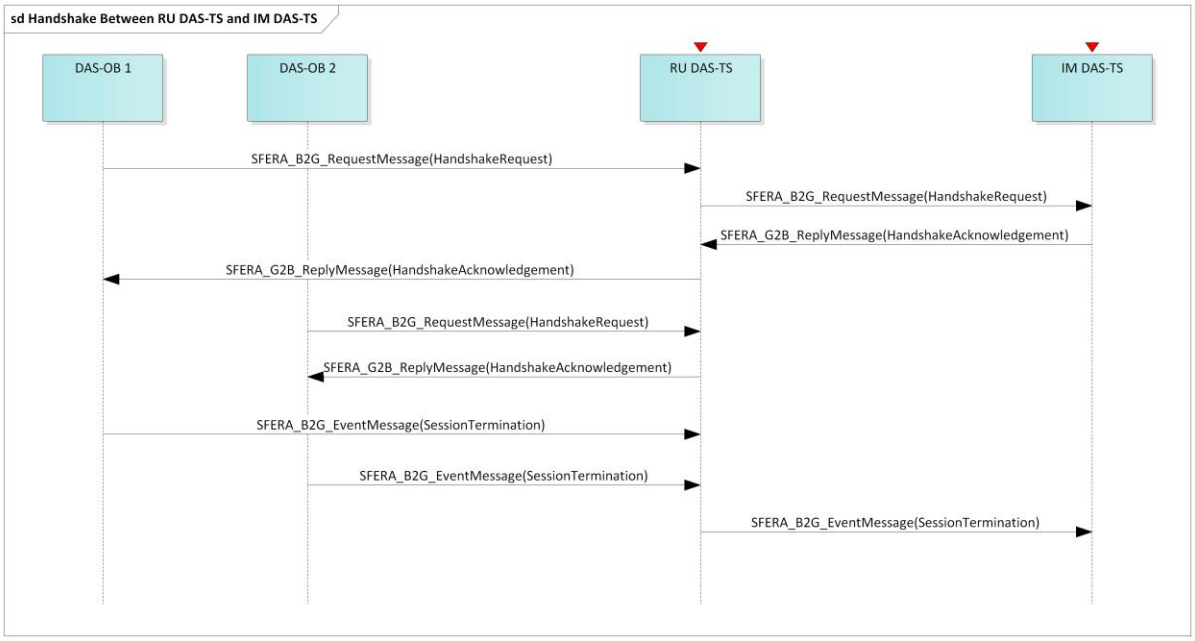


Figure 52: Sequence Diagram - Handshakes in IM-RU setup with multiple devices

The level of abstraction introduced means that the device identifiers (“SourceDevice” and “DestinationDevice” in the message header) are not useful in the RU DAS-TS/IM DAS-TS link. Therefore, they will not be included in the message header:

- for “G2B” messages, the RU DAS-TS is in charge of maintaining the association between Device Identifiers and Train Identifiers;
- for “B2G” messages, the RU filters the payloads relevant for the IM (e.g. only status reports coming from the driver’s device), and sends them without documenting the “SourceDevice” in the message header.